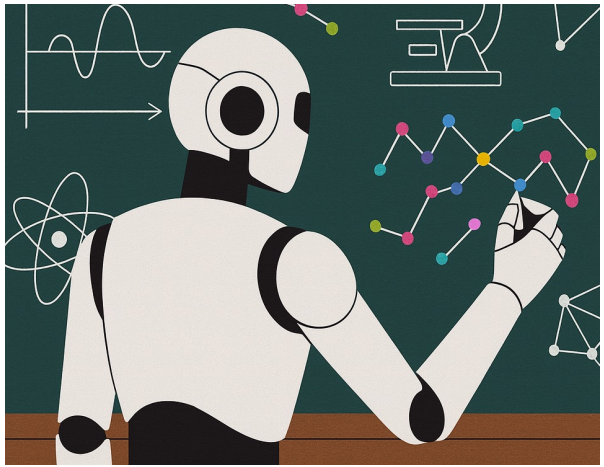
If you have any questions, feel free to reach out!

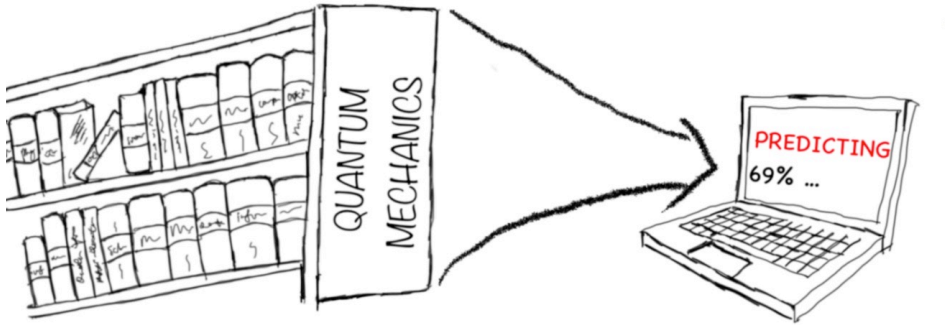
xuemei.gu@uni-jena.de

AI for Quantum Physics

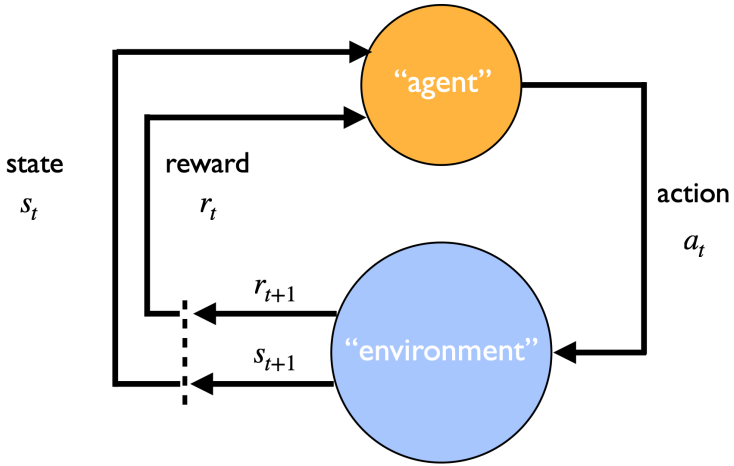


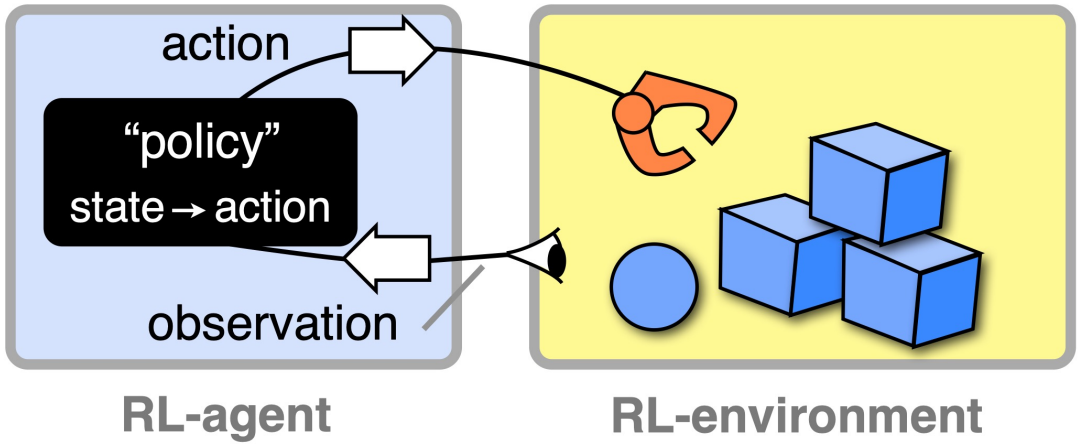
Winter semester 2025/26

Literature-based Scientific Discovery

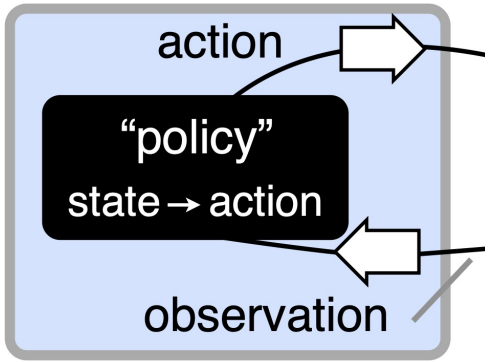


Recap of Reinforcement Learning





Policy — How does the agent behave?



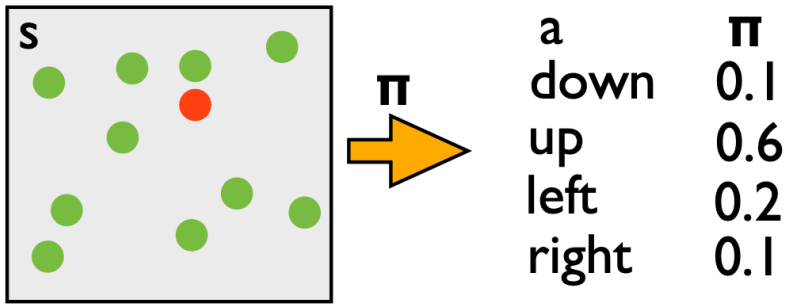
RL-agent

Policy: Mapping from states to probabilities of selecting each possible action

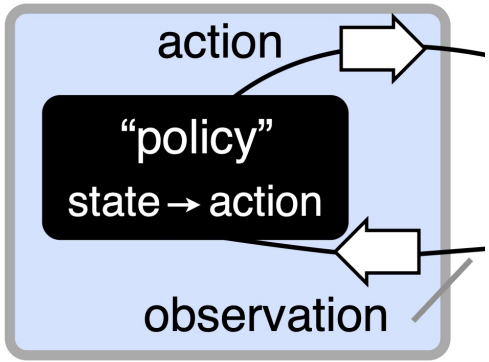
$$\pi_\theta(a_t \mid s_t)$$

Deterministic policy: $a = \pi(s)$

Stochastic policy: $\pi(a|s) = \mathbb{P}[A_t = a|S_t = s]$



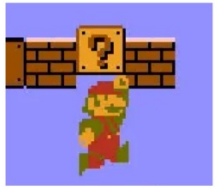
Experiences
Rewards



RL-agent

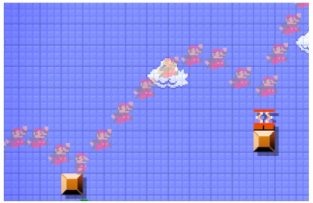
How good is a given state? $r_t = R(s_t)$

How good is a state-action pair? $r_t = R(s_t, a_t)$



How good is a *trajectory* $\tau = (s_0, a_0, s_1, a_1, \ldots)$:

$$R(\tau) = \sum_{t=0}^{\infty} \gamma^k r_t \text{ where } \gamma \in [0,1] \text{ is the temporal discount}$$



Value: Functions that estimate how good it is for the agent to be in a given state (assuming policy π_θ)

State-value function: Expected return when starting in s and following π_θ thereafter:

$$v_\pi(s) \equiv \mathbb{E}_\pi \left[\sum_{k=0} r_{t+k+1} \,|\, s_t = s \right]$$

Value Function

$$v_\pi(s) = \sum_a q_\pi(s,a)\pi_\theta(a \,|\, s)$$

Action-value function: Expected return when starting in s, taking action a, and following π_θ thereafter:

$$q_\pi(s,a) \equiv \mathbb{E}_\pi \left[\sum_{k=0} r_{t+k+1} \,|\, s_t = s,\ a_t = a \right]$$

Q-value Function

$$q_\pi(s,a) = \sum_{s',r} p(s', r \,|\, s, a)(r + v_\pi(s'))$$

Reinforcement Learning Approaches

Model-free RL

Value-based RL

- Estimate the optimal action-value function $Q^*(s, a)$
- No policy (implicit)

Policy-based RL

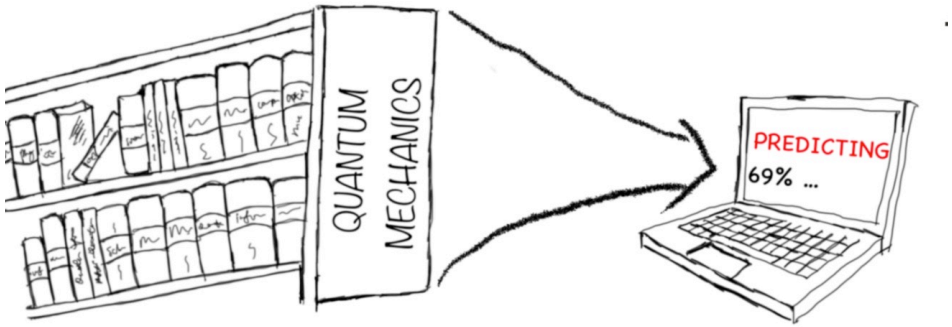
- Search directly for the optima policy π^*
- No value function

Actor-critic RL

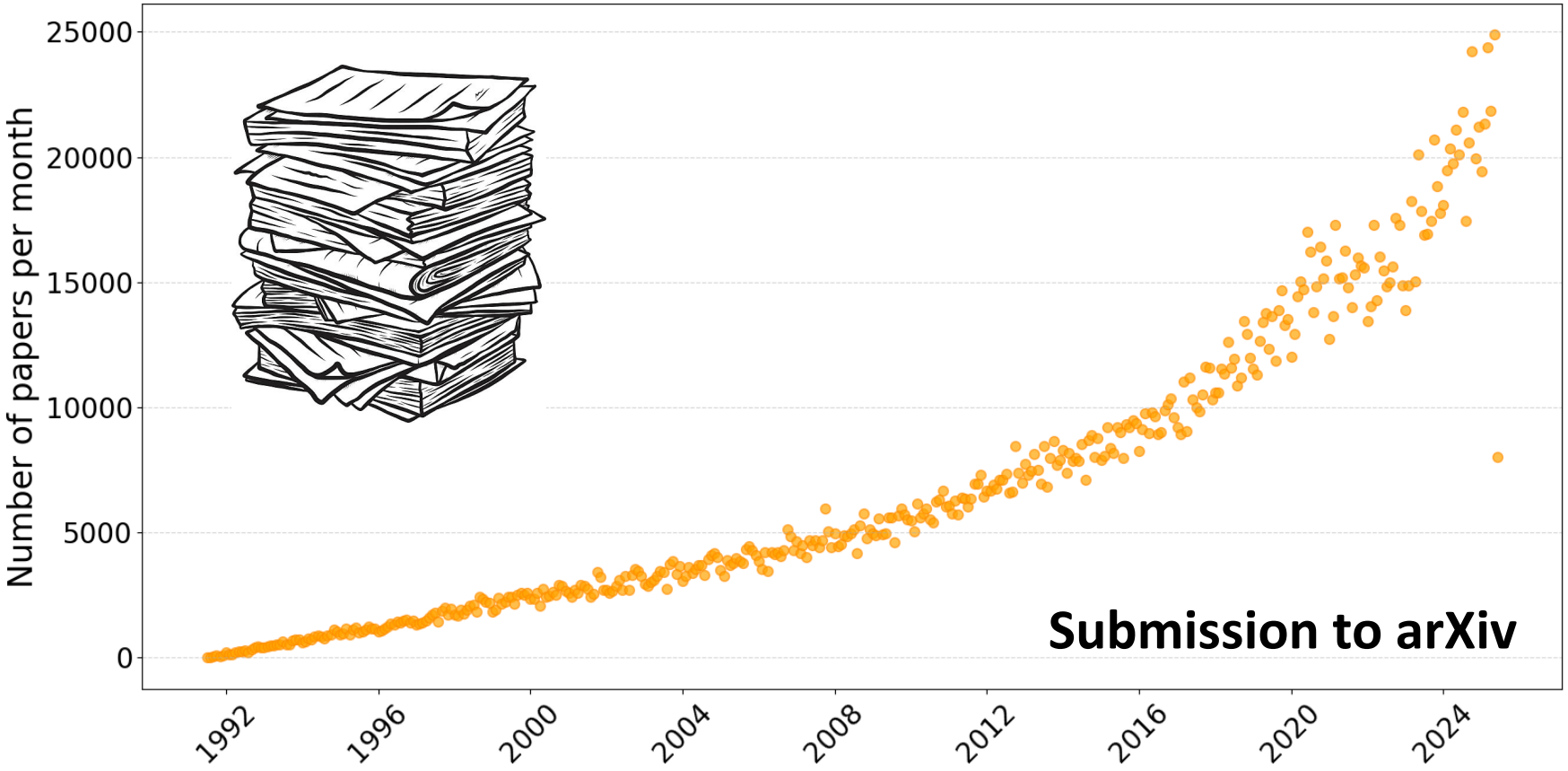
- Value function
- Policy function

* slide from Dhruv Batra

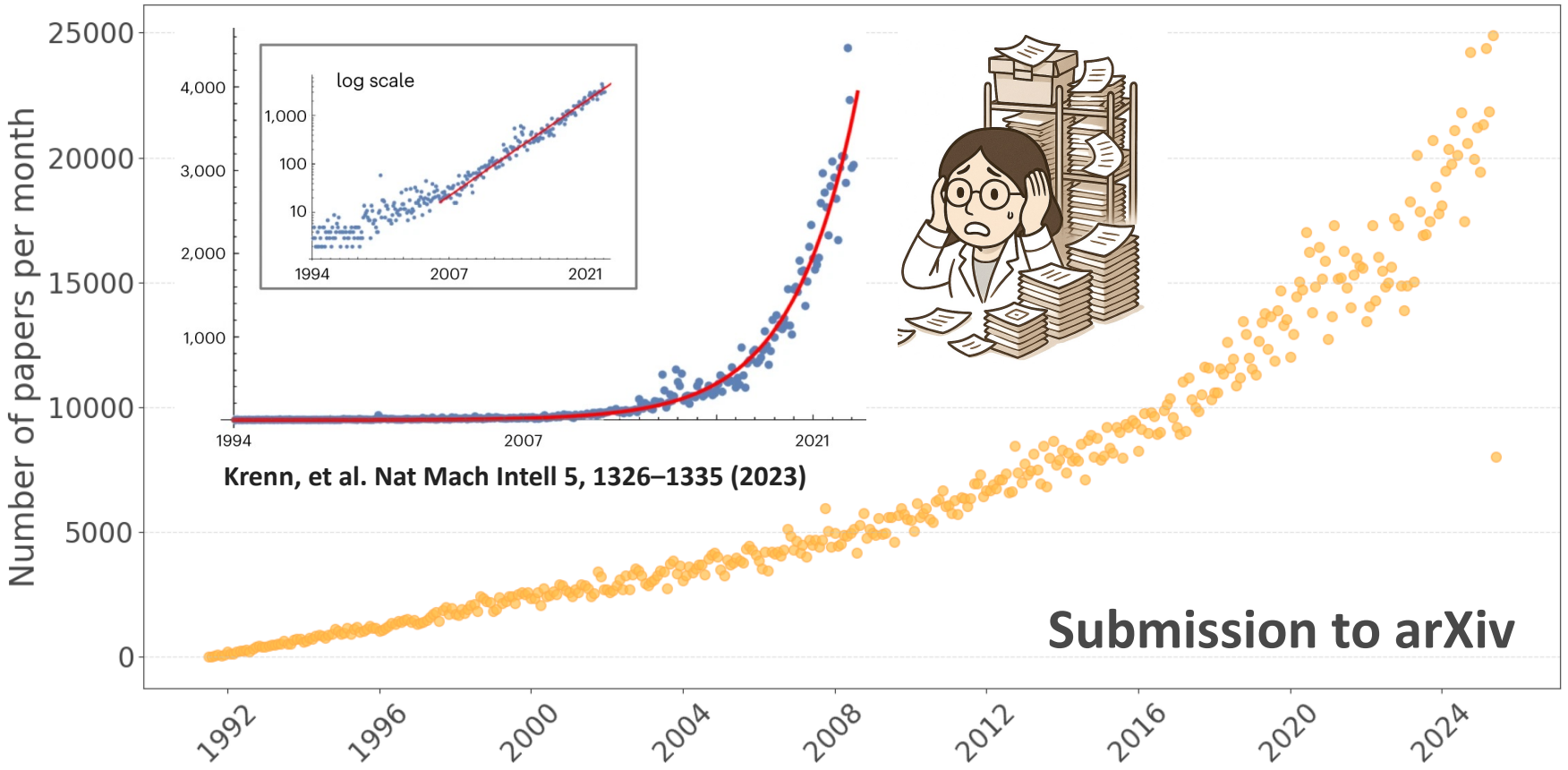
AI for Scientific Idea and Hypothesis Generation



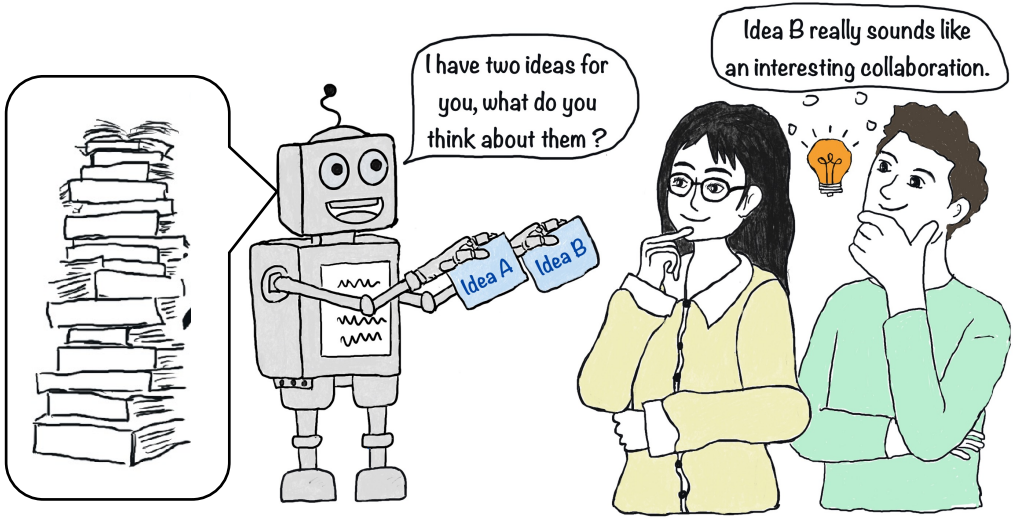
Overwhelming scientific knowledge



Overwhelming scientific knowledge



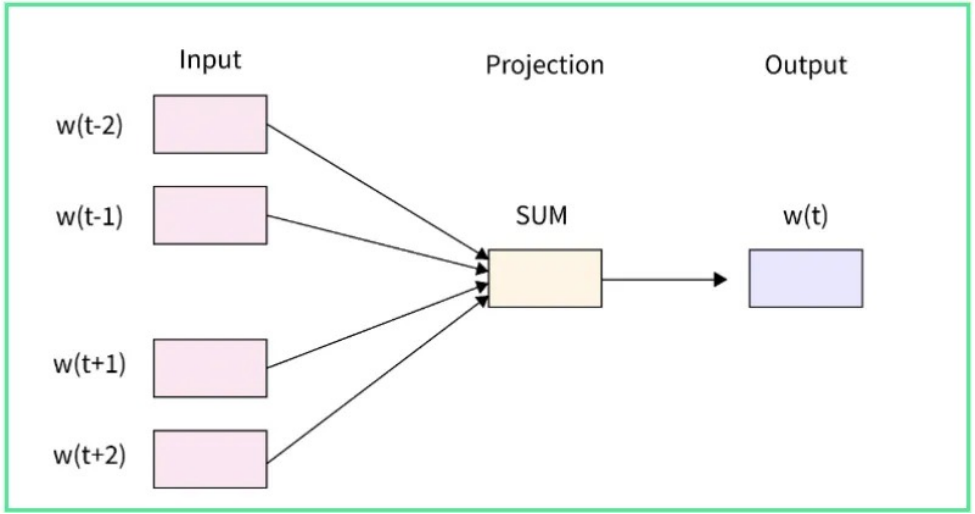
Tool that can inspire interesting/impactful ideas for scientists?



word embedding

e.g., represents words as dense, numerical vectors in a continuous multi-dimensional space, capturing their semantic meaning and relationships

Word2Vec: Continuous Bag of Words (CBOW)



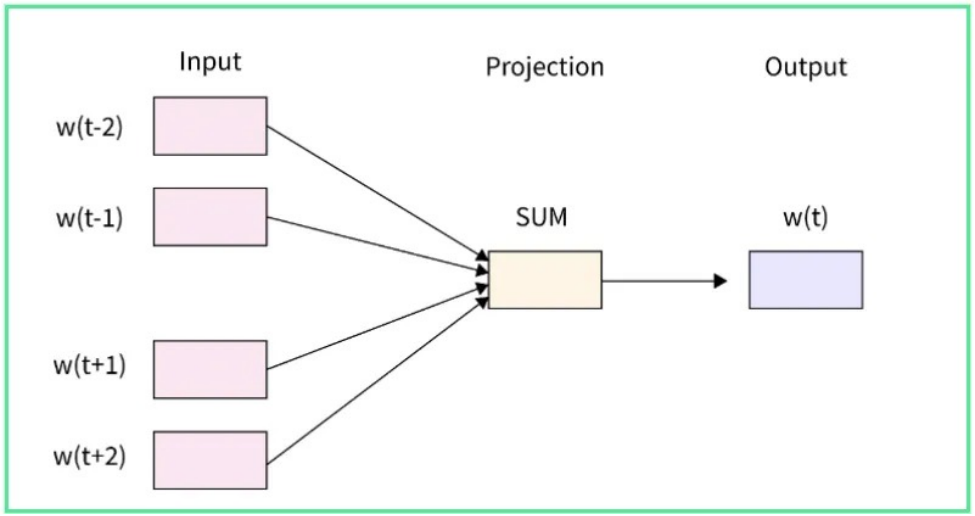

```python
from gensim.models import Word2Vec

sentences = [["quantum", "optics", "uses", "photons"]]

cbow = Word2Vec(
    sentences=sentences,
    vector_size=10,
    window=1,
    min_count=1,
    sg=0,         # CBOW
    epochs=10,
    workers=1,
    seed=0
)
```

Predict the center word from its surrounding context.

word embedding

e.g., represents words as dense, numerical vectors in a continuous multi-dimensional space, capturing their semantic meaning and relationships

Word2Vec: Skip-gram

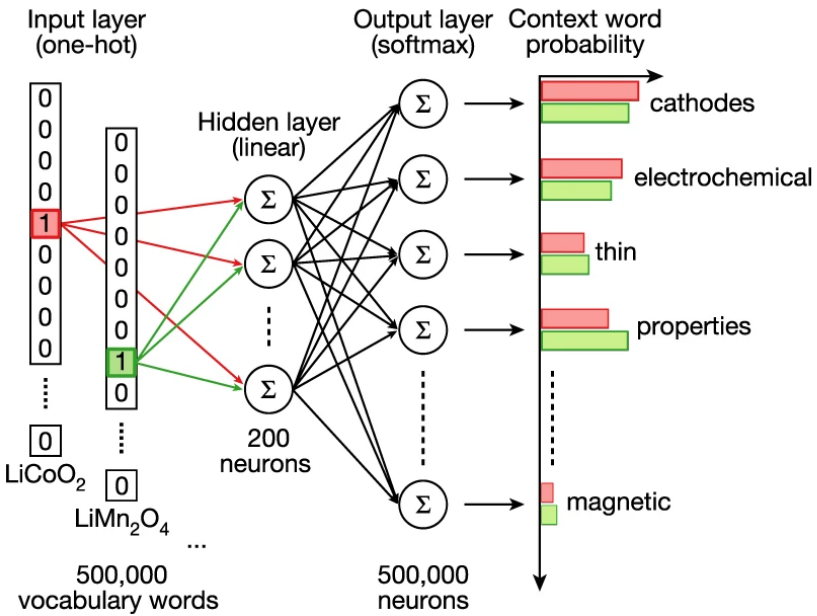


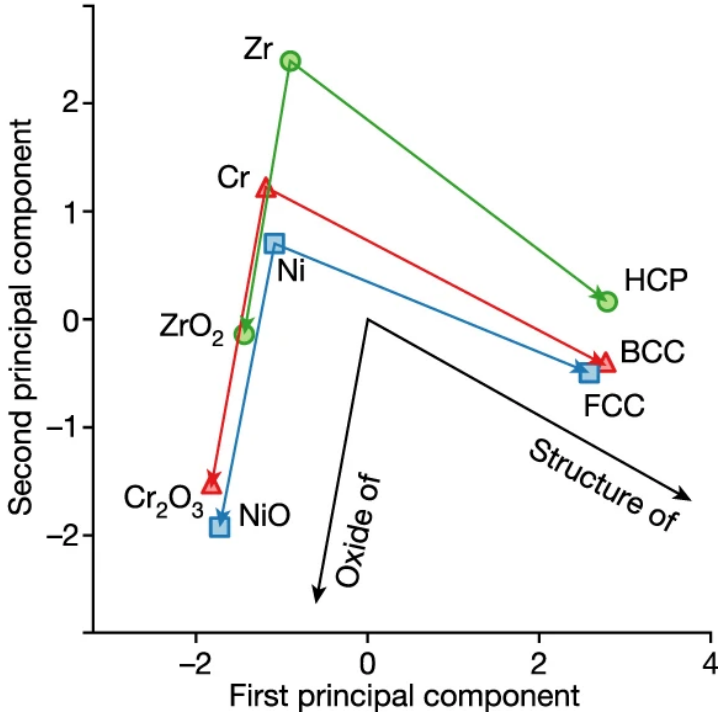

```python
from gensim.models import Word2Vec

sentences = [["quantum", "optics", "uses", "photons"]]

skipgram = Word2Vec(
    sentences=sentences,
    vector_size=10,
    window=1,
    min_count=1,
    sg=1,        # Skip-gram
    epochs=10,
    workers=1,
    seed=0
)
```

Predict surrounding context words from the center word.

Word2vec skip-gram and analogies

trained to predict all words mentioned within a certain distance (context words) from the given target word

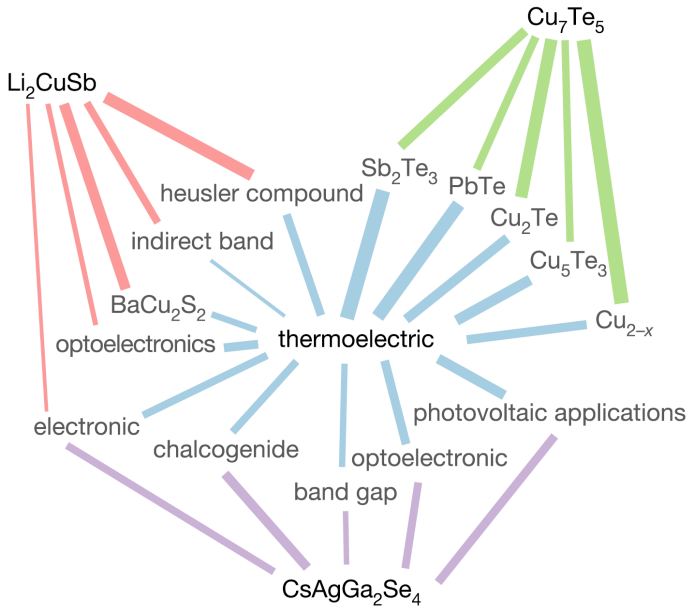




Tshitoyan, V., Dagdelen, J., Weston, L. et al. Unsupervised word embeddings capture latent knowledge from materials science literature. Nature 571, 95–98 (2019).

Word2Vec of 3,3mio papers

Prediction of new thermoelectric materials

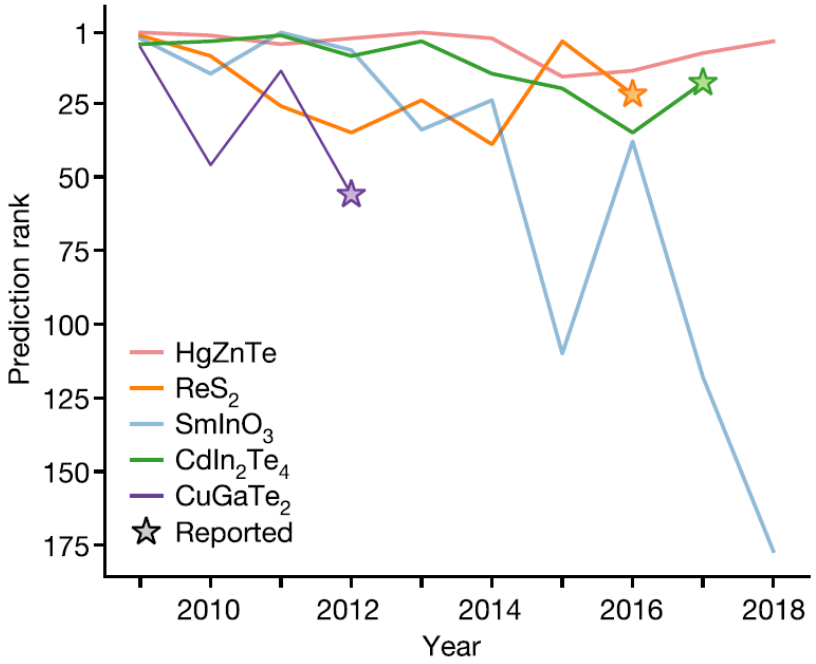


Cosine similarity to 'thermoelectric'

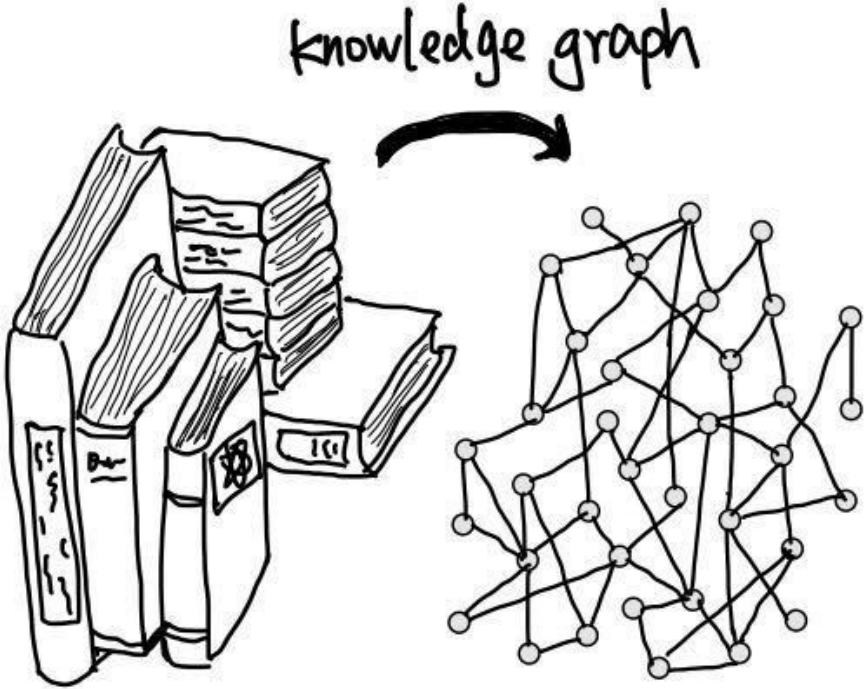
1.	Bi_2Te_3	✓
2.	MgAgSb	✓
3.	PbTe	✓
	...	✓
326.	Li_2CuSb	?
	...	✓
328.	In_4Te_3	✓
	...	✓
345.	$Cu_3Nb_2O_8$	?
	...	✓

✓ Known thermoelectric

? Predictions



Tshitoyan, V., Dagdelen, J., Weston, L. et al. Unsupervised word embeddings capture latent knowledge from materials science literature. Nature 571, 95–98 (2019).


knowledge graph

Choosing experiments to accelerate collective discovery

Andrey Rzhetsky[a,b,c,1], Jacob G. Foster[d], Ian T. Foster[b,e], and James A. Evans[b,f,1]

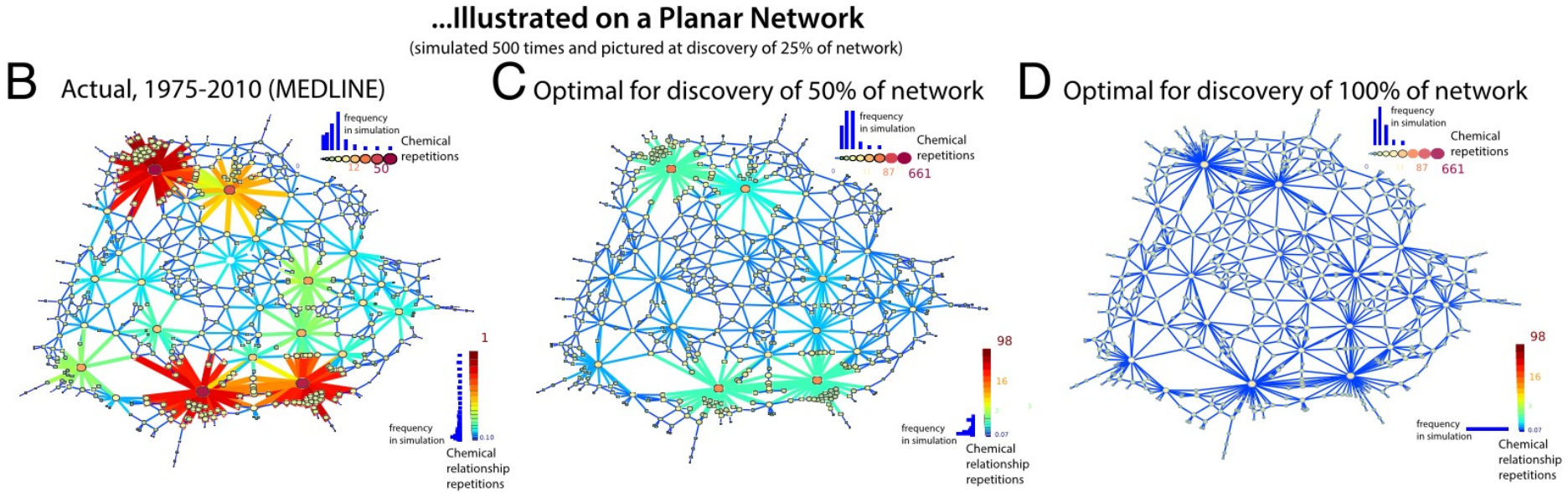
[a]Departments of Medicine and Human Genetics, University of Chicago, Chicago, IL 60637; [b]Computation Institute, University of Chicago and Argonne National Laboratory, Chicago, IL 60637; [c]Institute of Genomic and Systems Biology, University of Chicago, Chicago, IL 60637; [d]Department of Sociology, University of California, Los Angeles, CA 90095; [e]Mathematics and Computer Science Division, Argonne National Laboratory, Argonne, IL 60637; and [f]Department of Sociology, University of Chicago, Chicago, IL 60637

Edited by Yu Xie, University of Michigan, Ann Arbor, MI, and approved September 8, 2015 (received for review May 18, 2015)

How do scientists choose which experiments to do?
Are those choices good — overall — for the progress of science?

research problems as links between scientific entities in a knowledge network

Modeling Science as a Knowledge Network

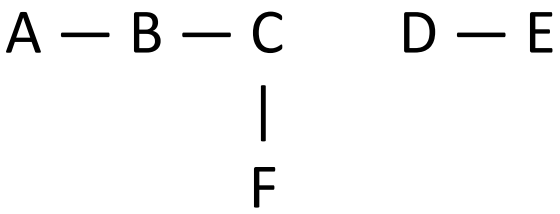


scientists tend to choose conservative, low-risk experiments that stay close to what's already known — and while that makes sense for individual careers, it slows down science as a whole.

Example:

A B C D E F

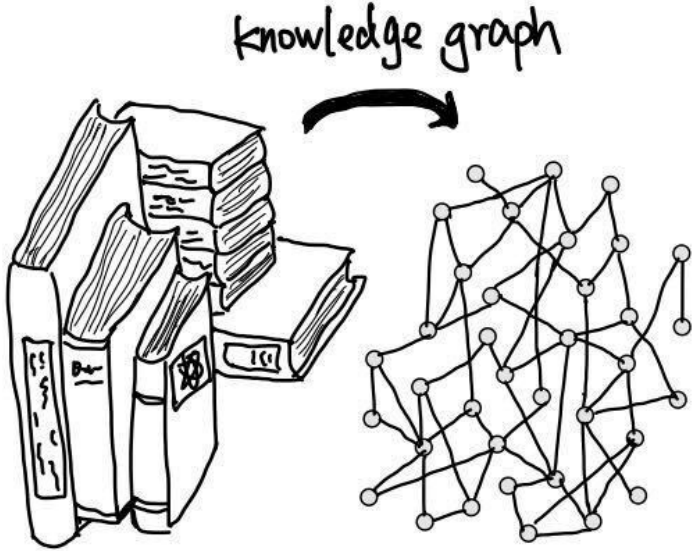
Conservative Strategy (What Scientists Often Do)



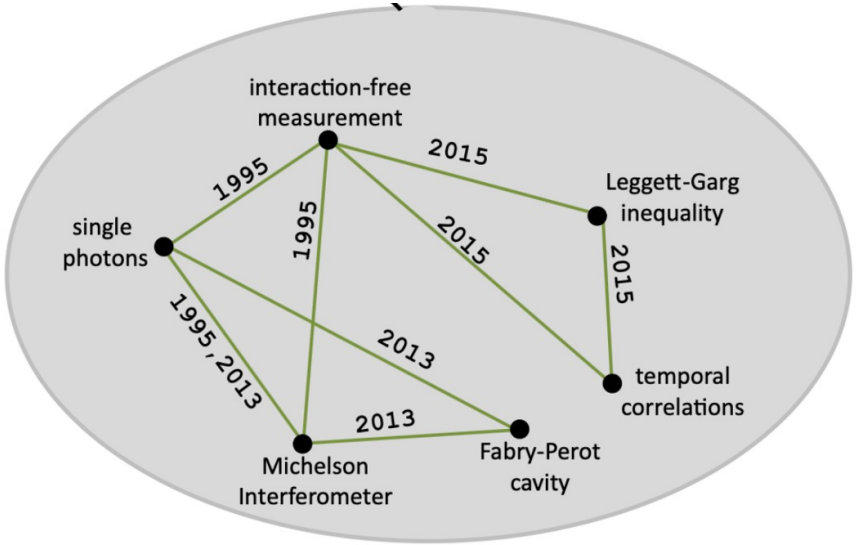
fragmented knowledge

Risky (Exploratory) Strategy

New Scientific Idea ?



M. Krenn, & A. Zeilinger, Predicting research trends with semantic and neural networks with an application in quantum physics, Proc. Natl. Acad. Sci. U.S.A. 117 (4) 1910-1916, https://doi.org/10.1073/pnas.1914370116 (2020).



Node: research concept
Edge: co-occurrence of concepts

Phys. Rev. Lett. 74, 4763 (1995):

[…] The fraction of **"interaction-free" measurements** can be arbitrarily close to 1. Using **single photons** in a **Michelson interferometer**, we have performed a preliminary demonstration of some of these ideas. […]

Phys. Rev. A 88, 023812 (2013):

We study the quantum dynamics of a **Michelson interferometer** with **Fabry-Perot cavity** arms and one movable end mirror, and driven by a **single photon**—an optomechanical device previously studied by Marshall et al. as a […]

Phys. Rev. X 5, 011003 (2015):

[…] Measuring **temporal correlations** of the position of single atoms performing a quantum walk, we observe a 6σ violation of the **Leggett-Garg inequality**. […] The **interaction-free measurement** is based on a novel atom transport system, […]

Rapid Automatic Keyword Extraction (RAKE)

"Bi2Te3 exhibits high Seebeck coefficient and low thermal conductivity, making it an efficient thermoelectric material."

1. Remove Stopwords & Punctuation

Stopwords: exhibits, and, it, an, the, of, is, making

Split text at stopwords → candidate phrases:

Bi2Te3
high Seebeck coefficient
low thermal conductivity
efficient thermoelectric material

2. Build Word Statistics

Frequency: how often a word appears
Degree: how many words it co-occurs with (including itself)

Word	Frequency	Degree
Bi2Te3	1	1
high	1	3
Seebeck	1	3
coefficient	1	3
low	1	3
thermal	1	3
conductivity	1	3
efficient	1	3
thermoelectric	1	3
material	1	3

Rapid Automatic Keyword Extraction (RAKE)

"Bi2Te3 exhibits high Seebeck coefficient and low thermal conductivity, making it an efficient thermoelectric material."

3. Score Each Word

$$score(word) = degree(word) / frequency(word)$$

Word	Frequency	Degree	Score (word)
Bi2Te3	1	1	1
high	1	3	3
Seebeck	1	3	3
coefficient	1	3	3
low	1	3	3
thermal	1	3	3
conductivity	1	3	3
efficient	1	3	3
thermoelectric	1	3	3
material	1	3	3

4. Score Each Phrase

$$score(phrase) = \sum score(words\ in\ phrase)$$

phrase	score
Bi2Te3	1
high Seebeck coefficient	9
low thermal conductivity	9
efficient thermoelectric material	9

5. Rank Output

Rapid Automatic Keyword Extraction (RAKE)

an unsupervised, domain-independent, and language-independent algorithm
for extracting key phrases from individual documents

Goal: Extract the most important **keyword phrases** from a single document

Algorithm (High-Level Steps)

1. **Remove** stopwords & punctuation → Chop text into candidate phrases

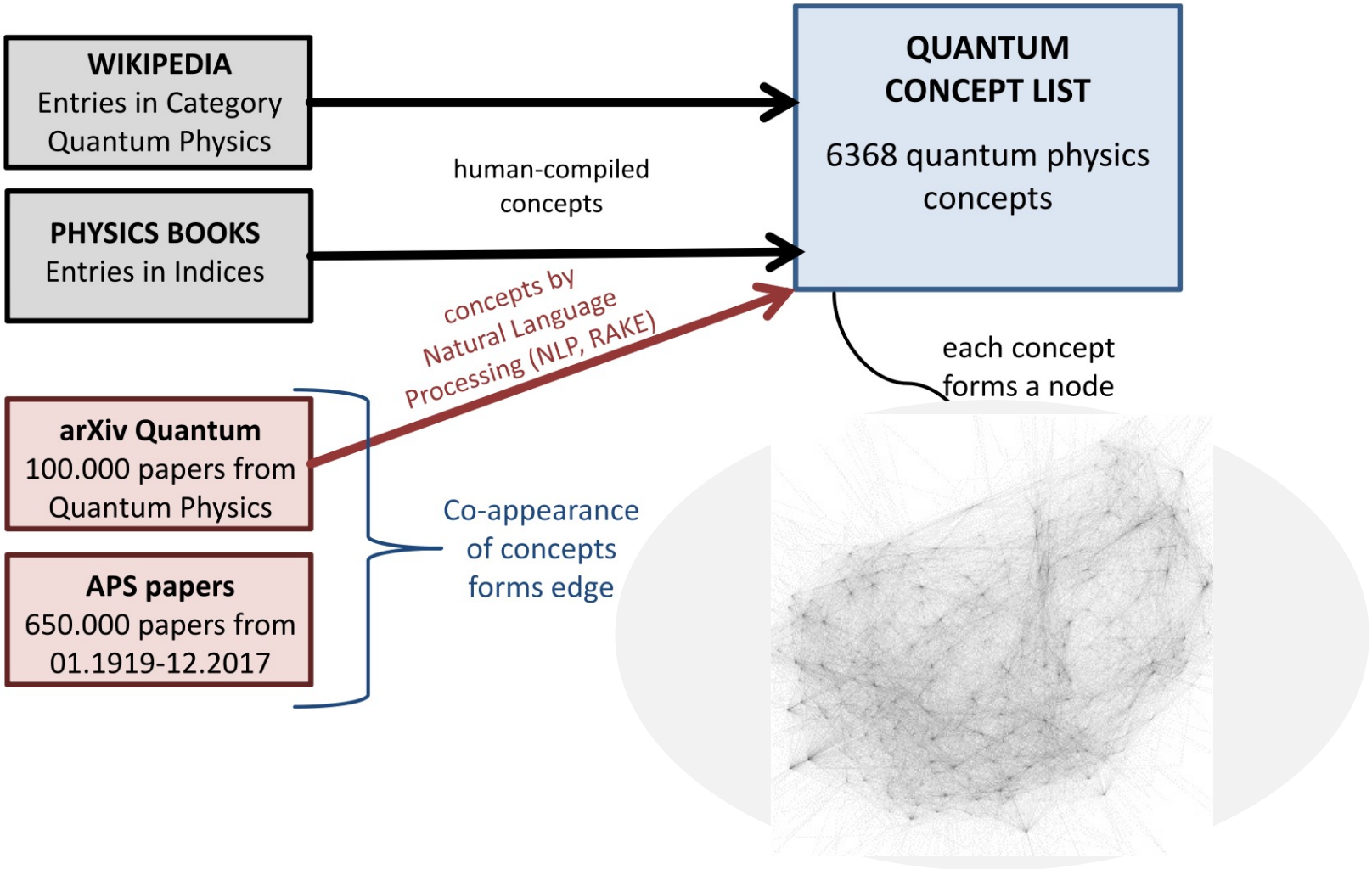
   "high Seebeck coefficient", *"low thermal conductivity"*

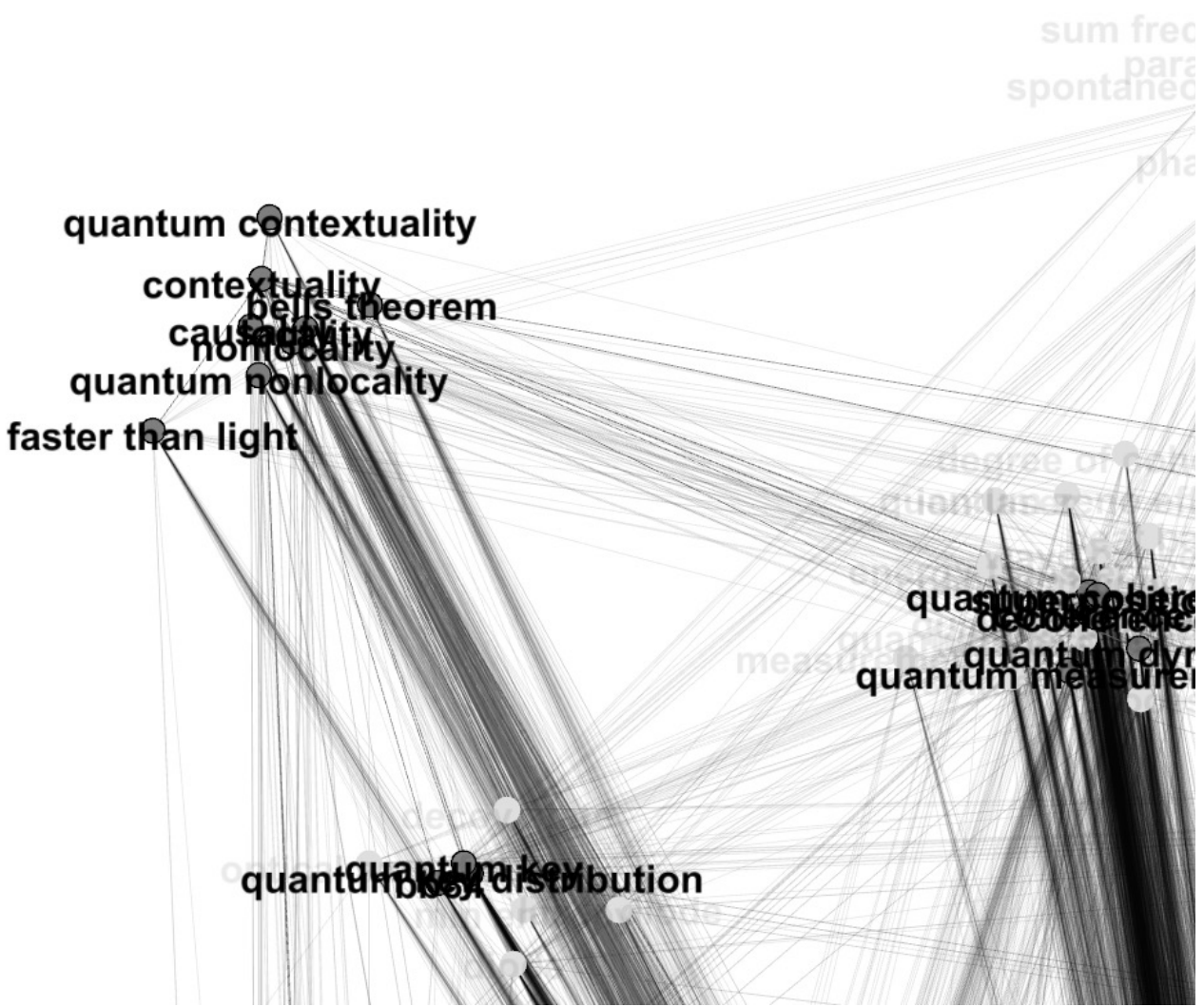
2. **Build word statistics**
   - ✔ ***Frequency:*** how often a word appears
   - ✔ ***Degree:*** how many words it co-occurs with

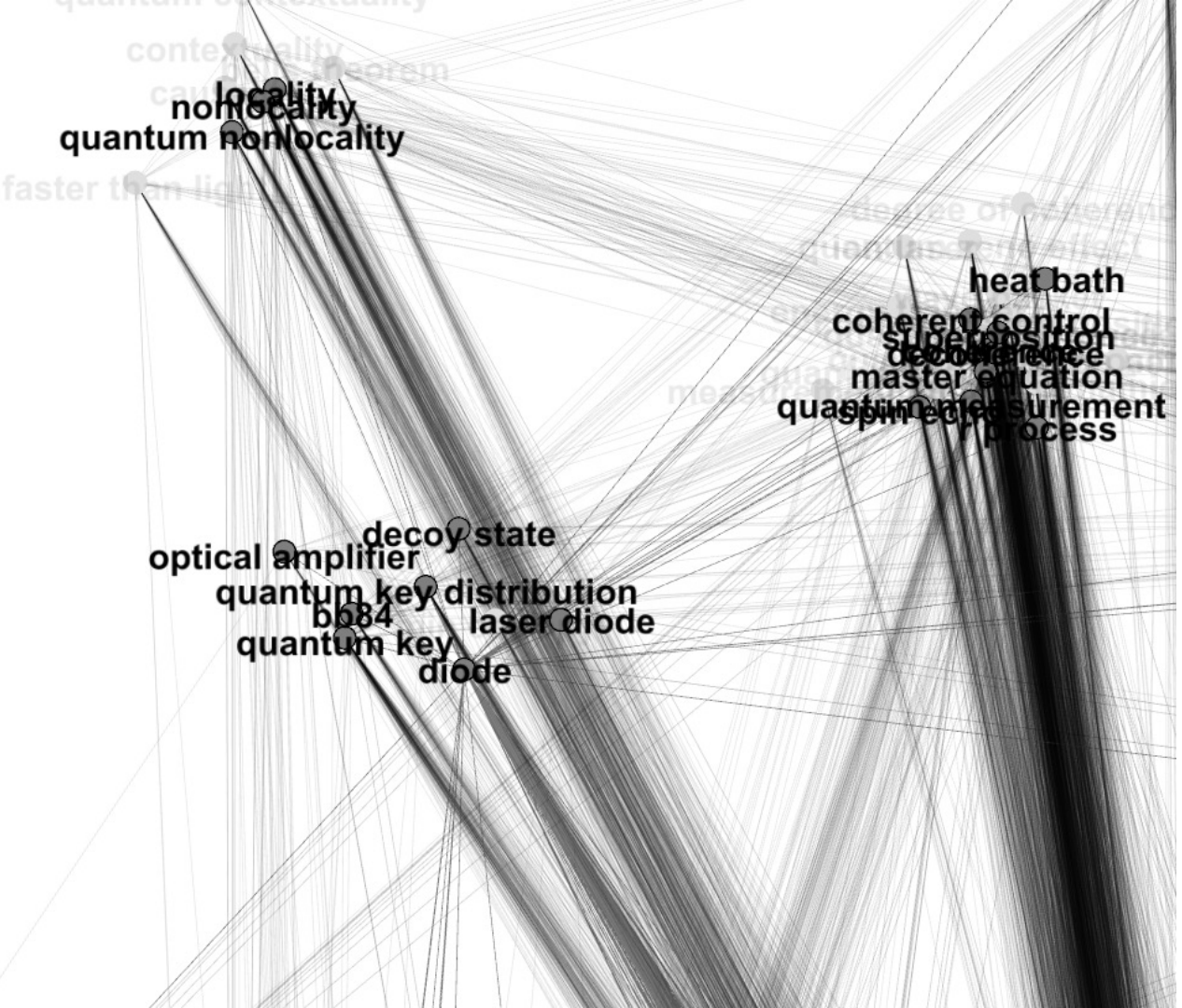
3. **Score words:** $score(word) = degree(word) / frequency(word)$

4. **Score phrases:** $score(phrase) = \sum score(words\ in\ phrase)$

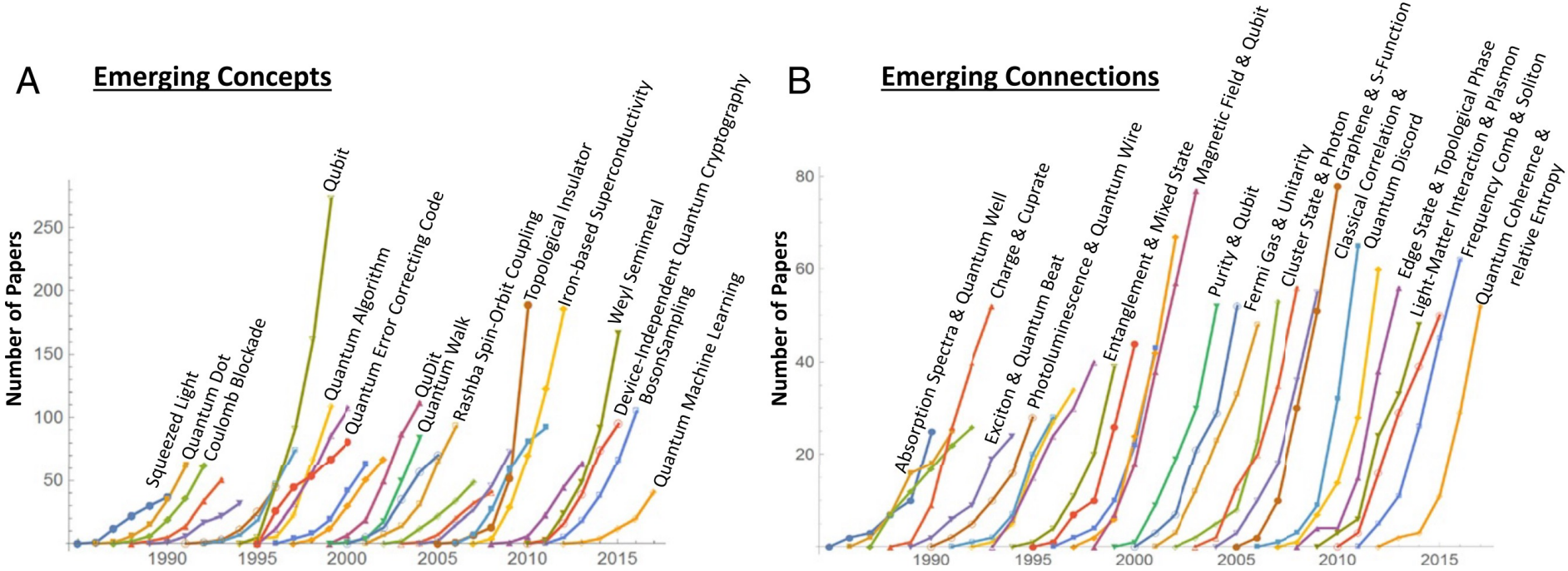
5. **Rank phrases** → Top-ranked phrases = **Keywords**

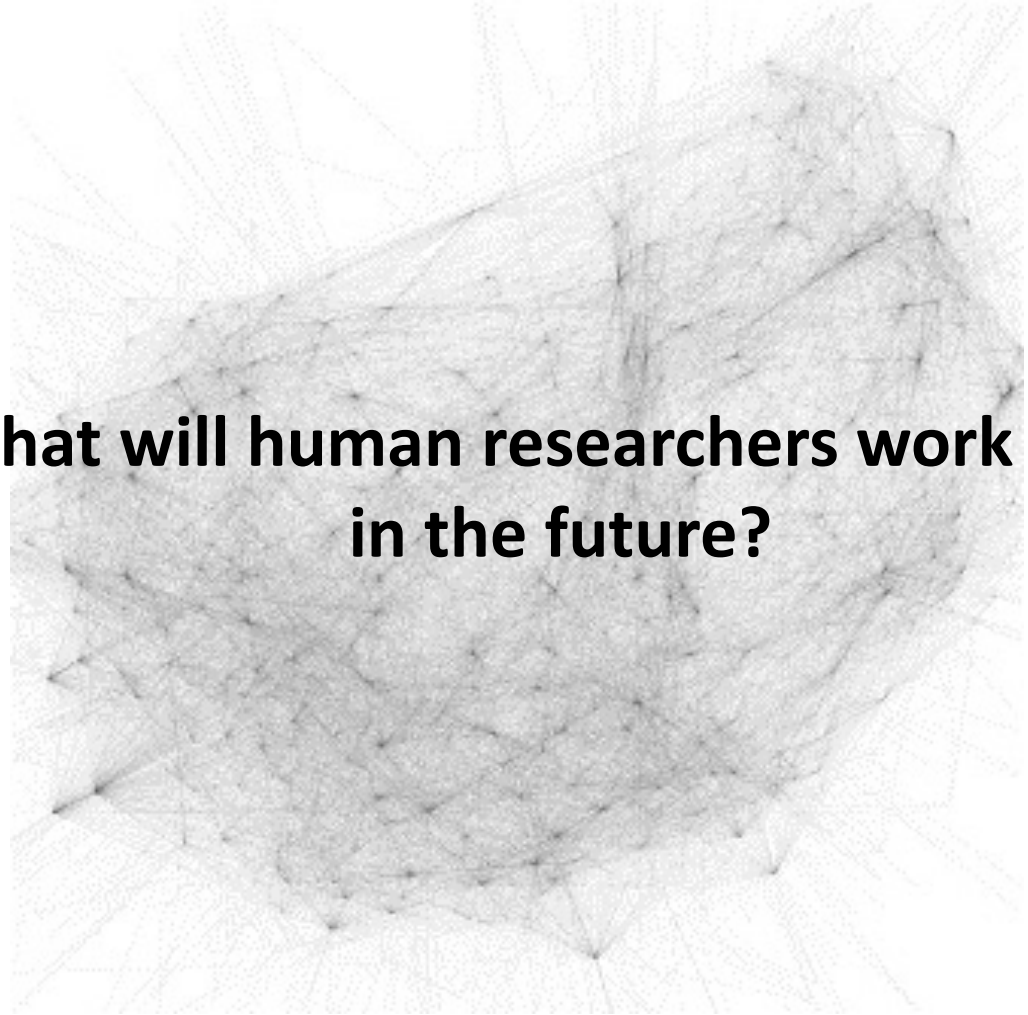

WIKIPEDIA
Entries in Category Quantum Physics
PHYSICS BOOKS
Entries in Indices
human-compiled concepts
QUANTUM CONCEPT LIST
6368 quantum physics concepts
concepts by Natural Language Processing (NLP, RAKE)
each concept forms a node
arXiv Quantum
100.000 papers from Quantum Physics
APS papers
650.000 papers from 01.1919-12.2017
Co-appearance of concepts forms edge


quantum contextuality
contextuality
bells theorem
nonlocality
quantum nonlocality
faster than light
quantum key
distribution


locality
nonlocality
quantum nonlocality
heat bath
coherent control
master equation
process
decoy state
optical amplifier
quantum key distribution
bb84
laser diode
quantum key
diode

Some History of Quantum Physics

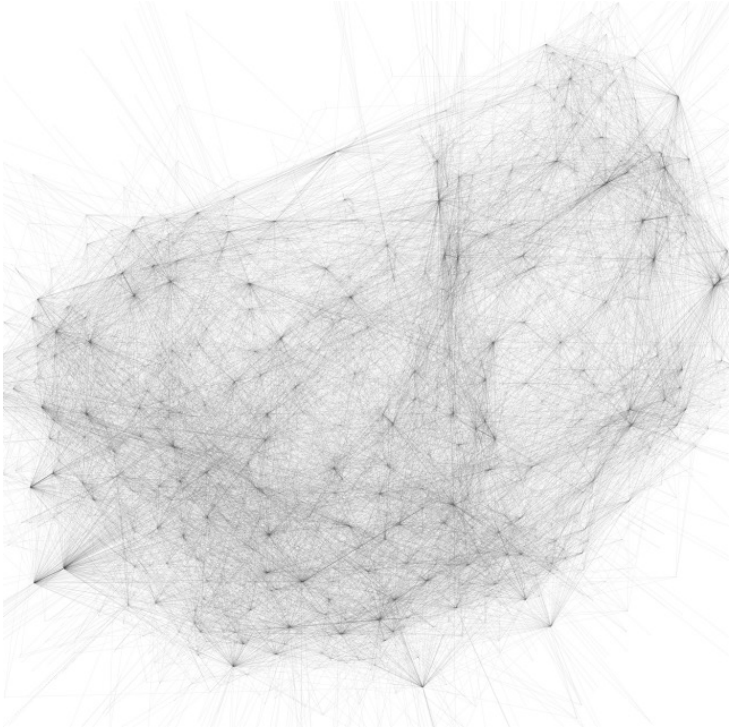




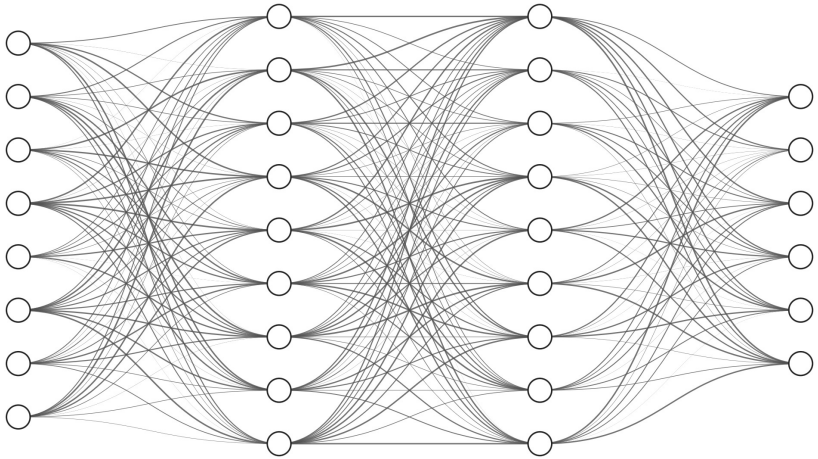
What will human researchers work on in the future?

Predicting the future of Quantum Physics

Which unconnected concept pairs will be investigated together in 5 years?

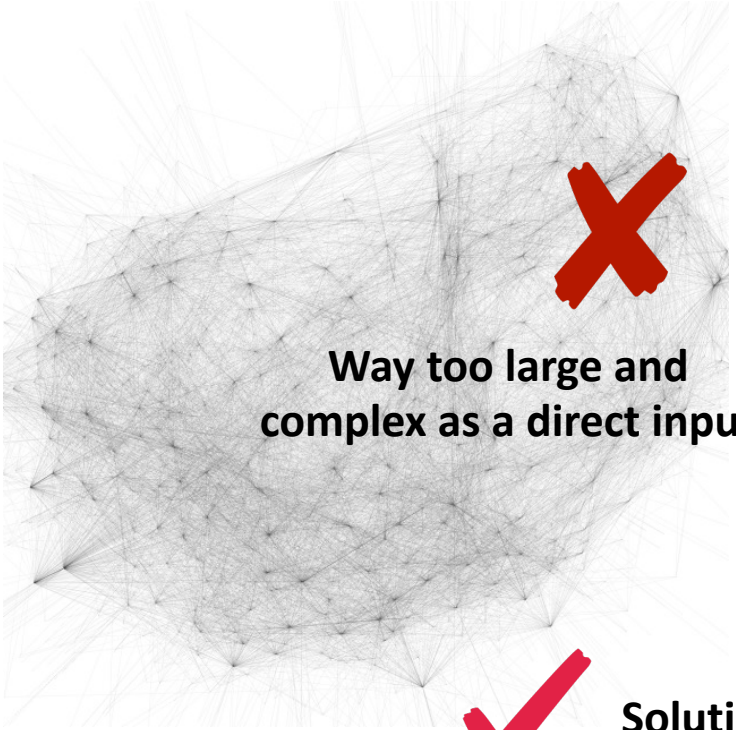


Neural Network



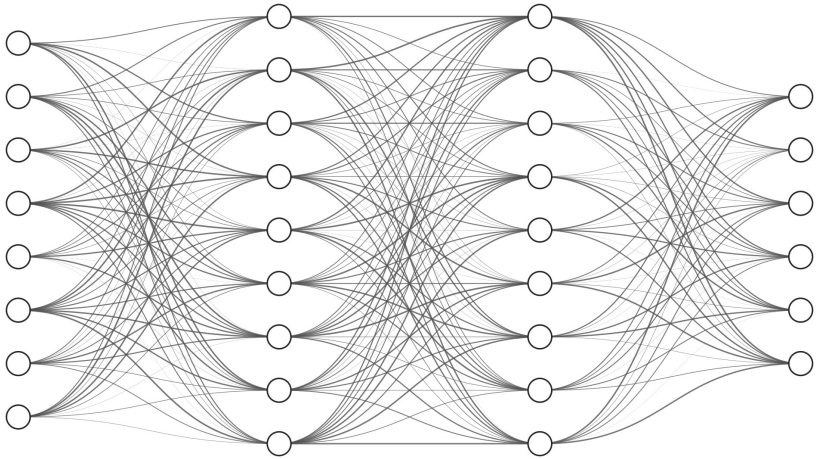
Predicting the future of Quantum Physics

Which unconnected concept pairs will be investigated together in 5 years?



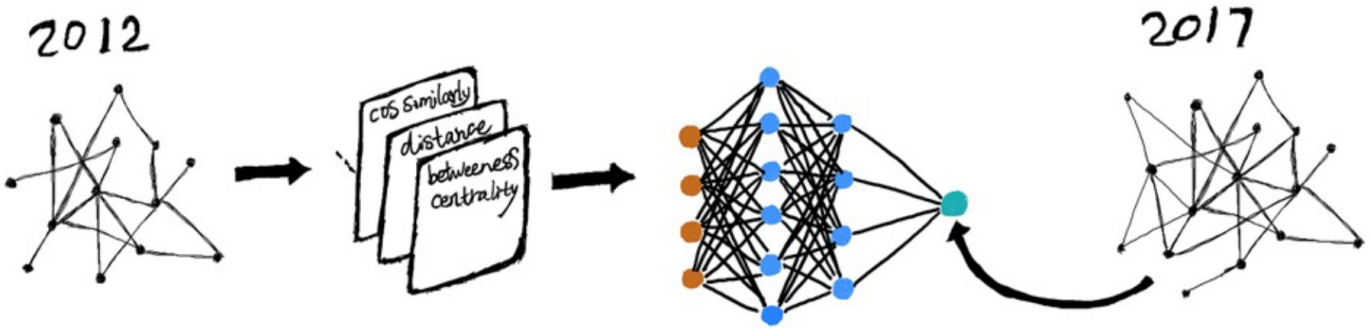
Way too large and complex as a direct input

Neural Network



Solution: Network Properties of pairs of concepts

Link Prediction via Neural Networks



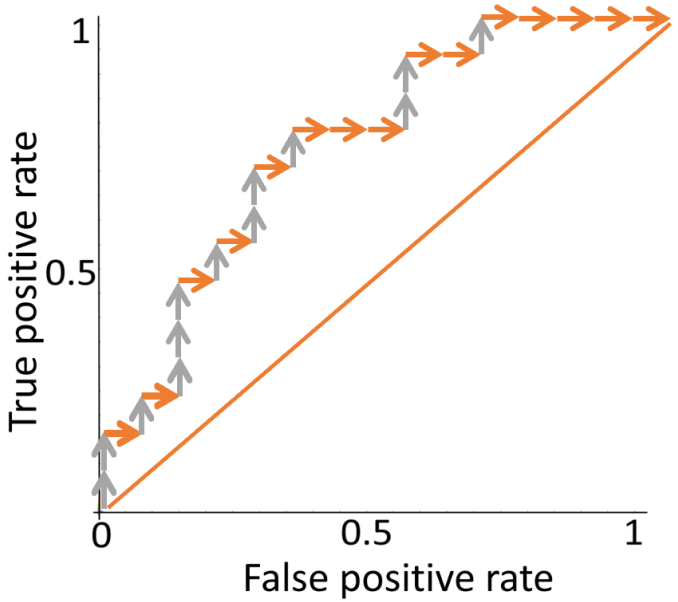
For each concept pair:

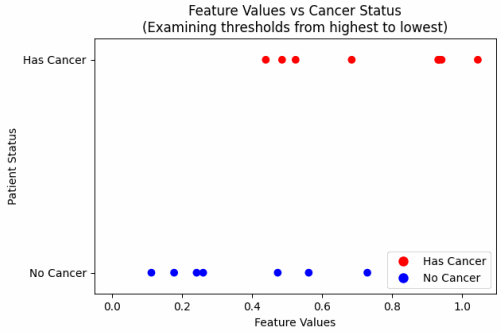
- #(papers of concept)
- #(connections of concept)
- #(shared Neighbors)
- Network Distances
- #(Paths of Length 2, 3, 4 over last 3 years)

```
class ff_network(nn.Module):

    def __init__(self):
        """
        Fully Connected layers
        """
        super(ff_network, self).__init__()

        self.semnet = nn.Sequential(
            nn.Linear(17, 50), # 17 properties
            nn.ReLU(),
            nn.Linear(50, 10), # very small network for tests
            nn.ReLU(),
            nn.Linear(10, 1)
        )


    def forward(self, x):
        """
        Pass throught network
        """
        res = self.semnet(x)

        return res
```

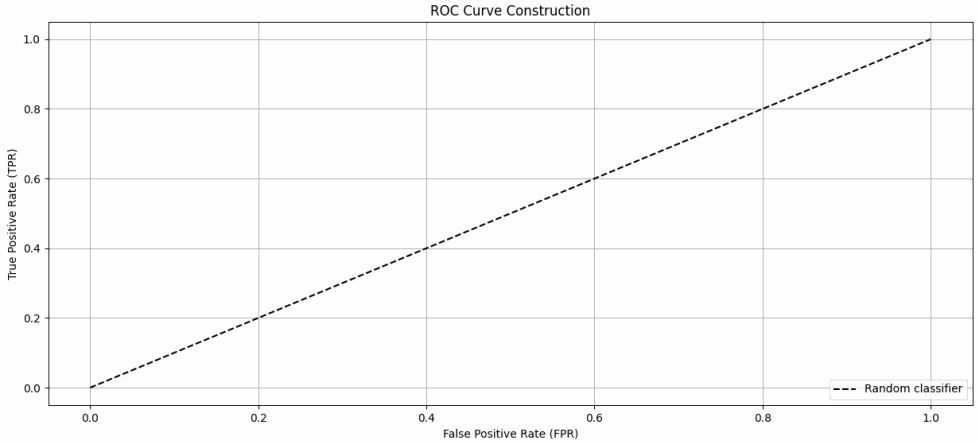
ROC and AUC

Quantify Quality of Prediction - ROC:
(Receiver Operating Characteristic)



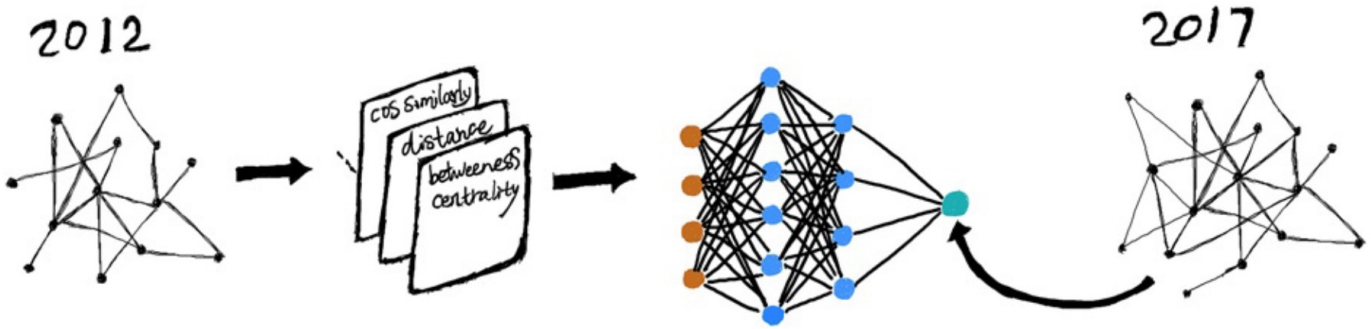


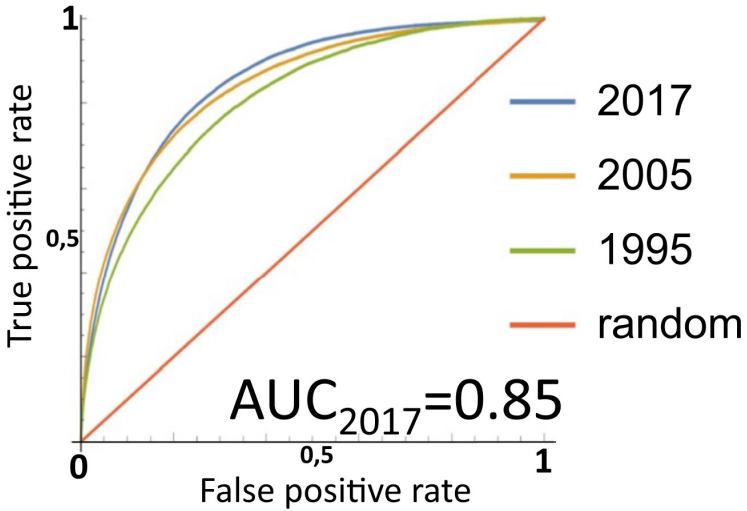
Current Metrics



https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc

Link Prediction via Neural Networks

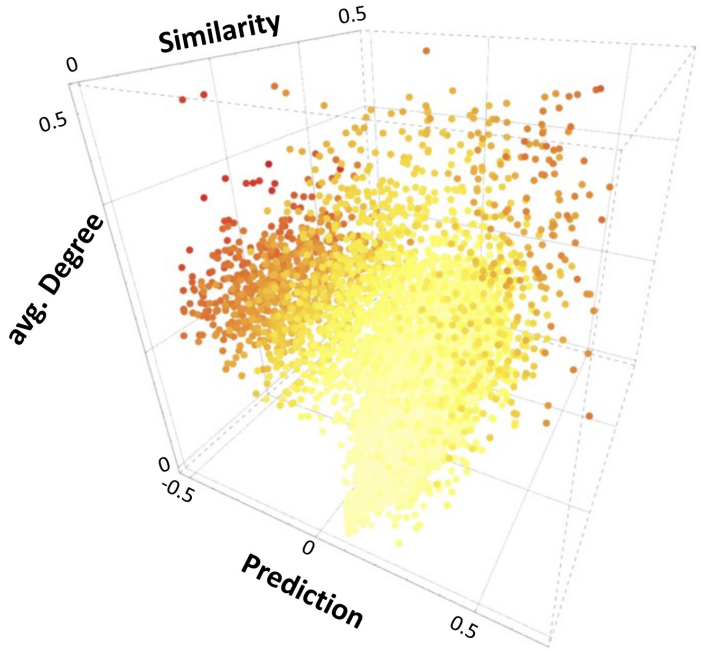




2017 → 2022
2022 → 2027

Personalized Suggestions

Prediction-Degree-Similarity of
100.000 personalized Concept Pairs



Every dot: Pair of quantum physics concepts

Concepts used in Researcher A's paper
with other physical concepts - What is predicted?

Predicted:

1: hybrid system, classical communication: (cosS: 0.30407, deg: 0.22924, pred: 1)
2: excited state, stanene: (cosS: 0.15908, deg: 0.35028, pred: 0.85577)
3: circuit qed, gravitational wave detection: (cosS: 0.34178, deg: 0.19095, pred: 0.85368)
4: optical transition, machine learning: (cosS: 0.29051, deg: 0.20901, pred: 0.84727)

Unorthodox (small cosS):

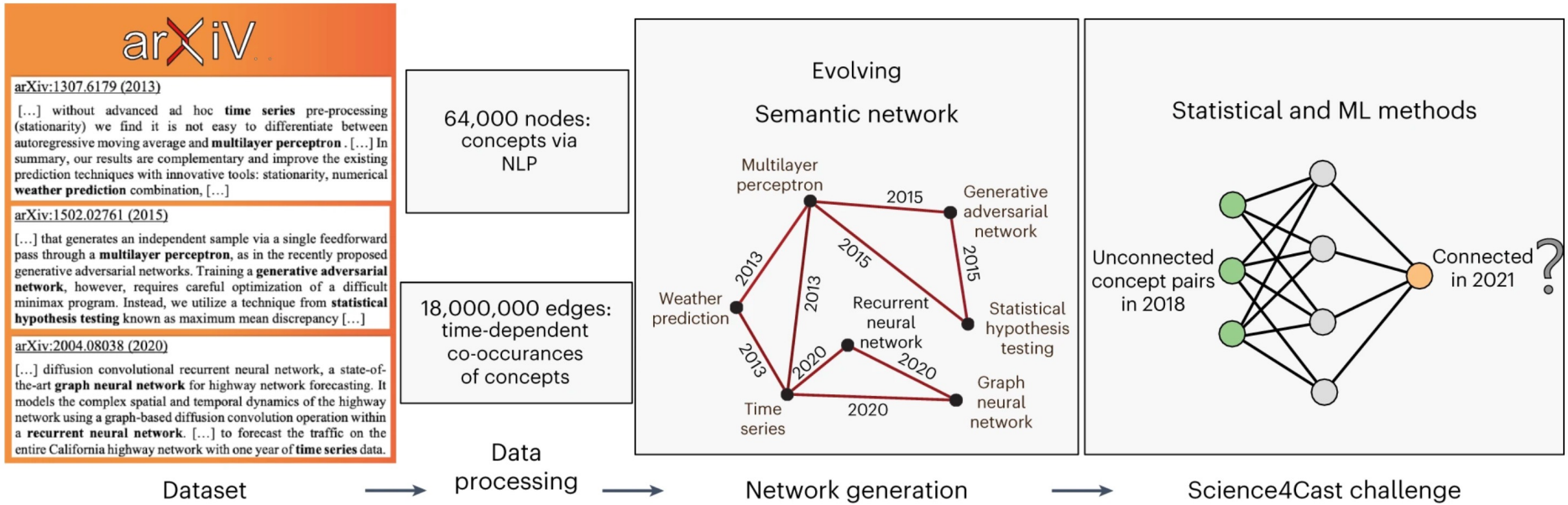
1: wave function, stanene: cosS: 0.14554, deg: 0.41675, pred: 0.85192
2: ground state, laser printing: cosS: 0.080176, deg: 0.43108, pred: 0.79129
3: optical lattice, electromagnetically induced grating: cosS: 0.12275, deg: 0.24917, pred: 0.71311

Unpredicted:

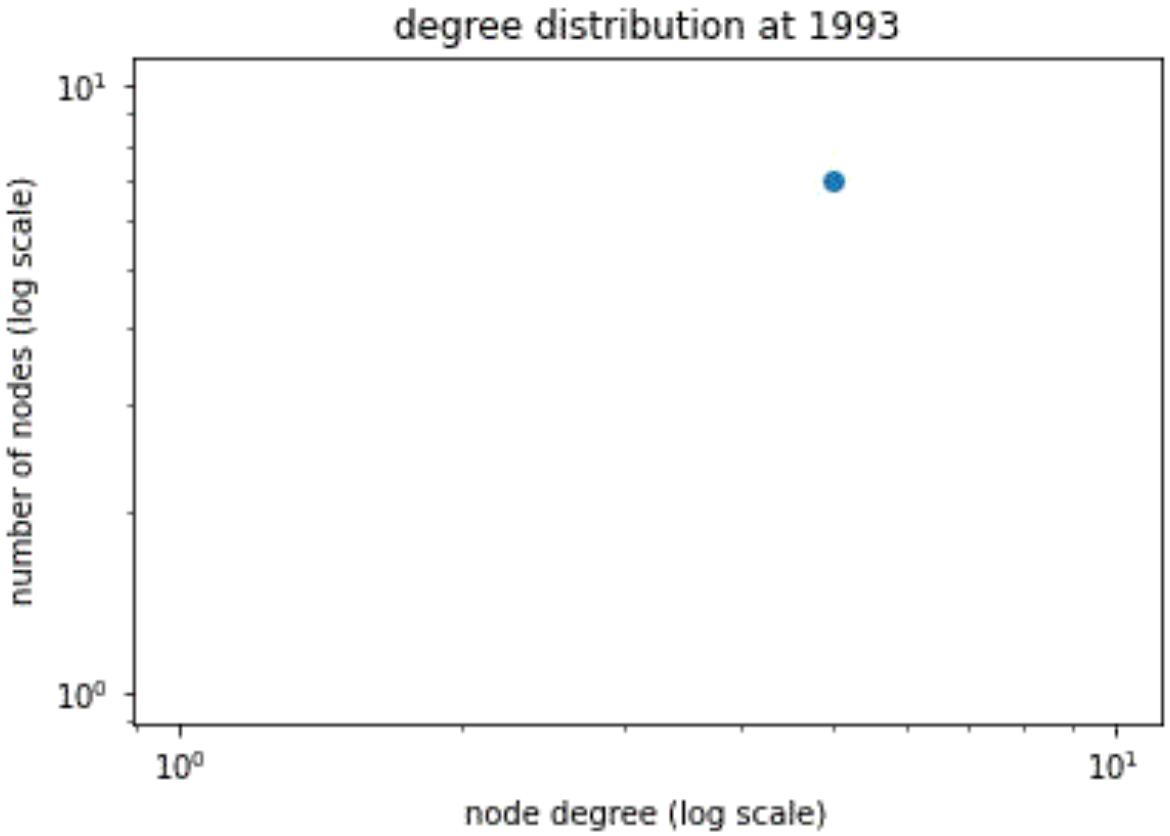
1: langevin equation, pseudogap: cosS: 0.48992, deg: 0.24897, pred: -0.99167
2: thermodynamic limit, multiple scattering: cosS: 0.53212, deg: 0.35657, pred: -0.95967
3: topological phase, molecular beam: cosS: 0.46109, deg: 0.25888, pred: -0.95818

1st AUC: ~92.8%

diverse techs: GNNs, Transformers, auto-feature extraction, pure statistical solutions, …

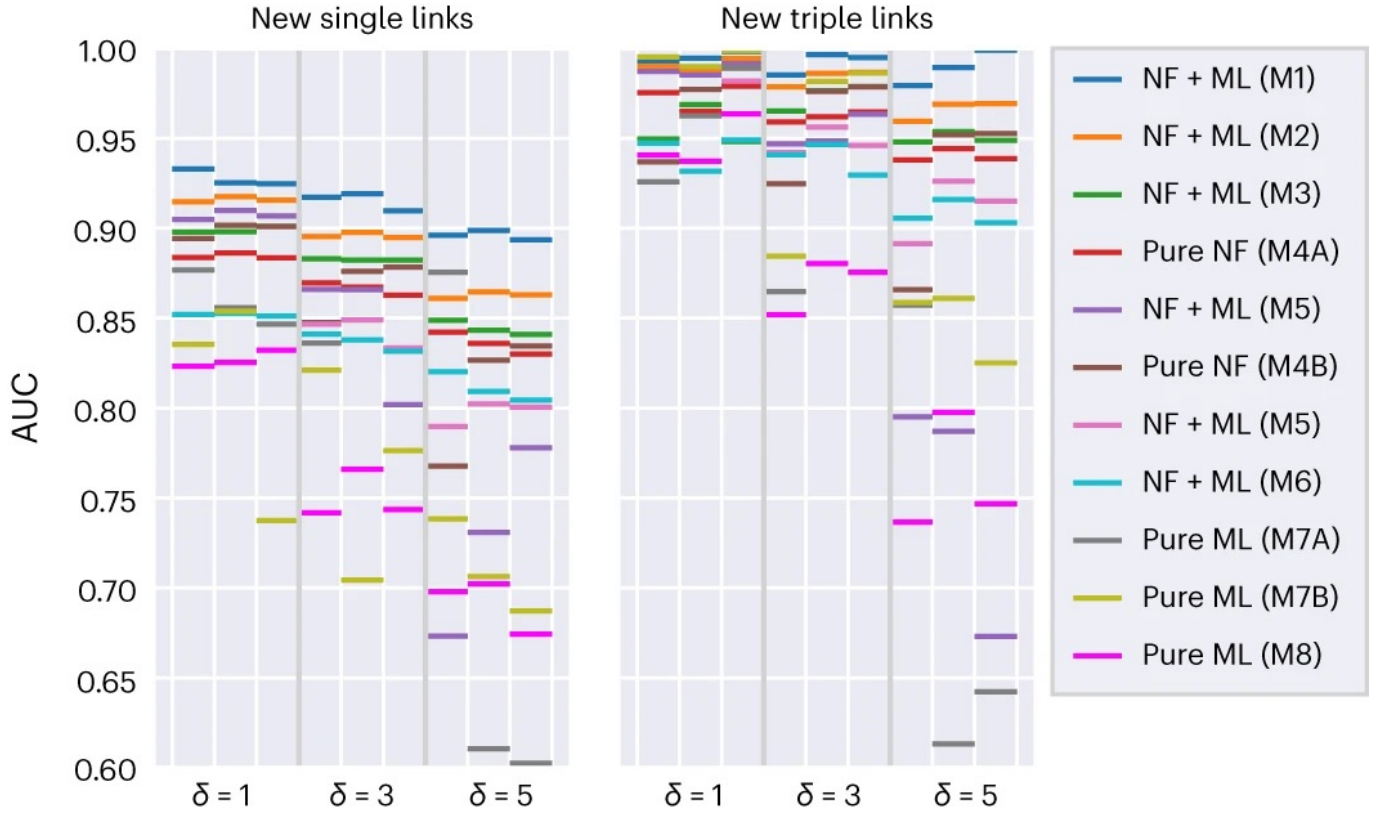


Krenn, M., Buffoni, L., Coutinho, B. et al. Forecasting the future of artificial intelligence with machine learning-based link prediction in an exponentially growing knowledge network.Nat Mach Intell 5, 1326–1335 (2023). https://doi.org/10.1038/s42256-023-00735-0



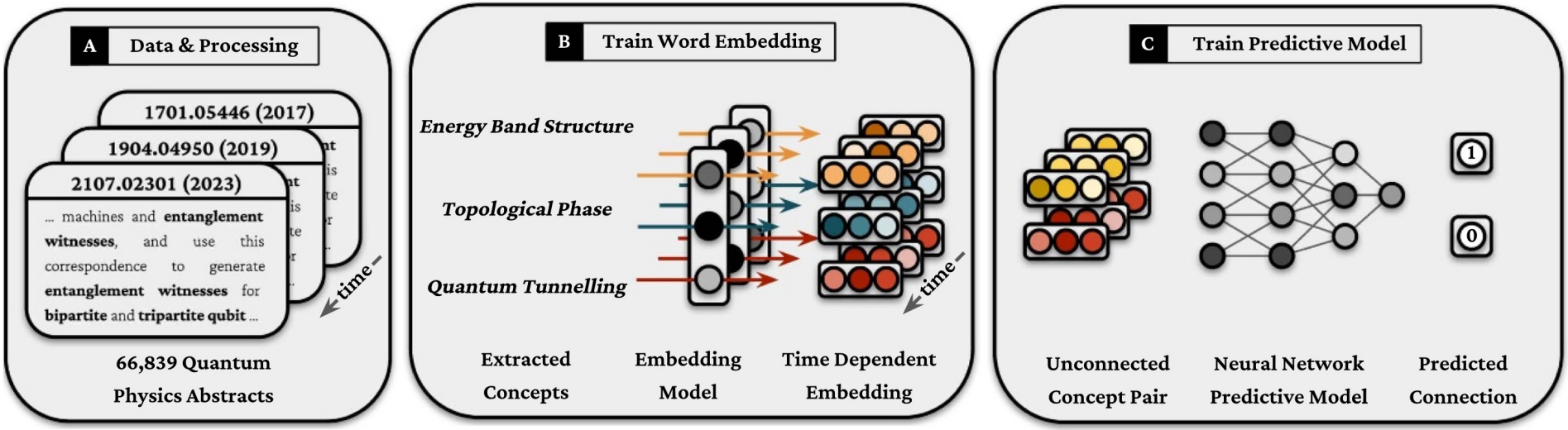
Krenn, M., Buffoni, L., Coutinho, B. et al. Forecasting the future of artificial intelligence with machine learning-based link prediction in an exponentially growing knowledge network.Nat Mach Intell 5, 1326–1335 (2023). https://doi.org/10.1038/s42256-023-00735-0

https://github.com/artificial-scientist-lab/FutureOfAIviaAI

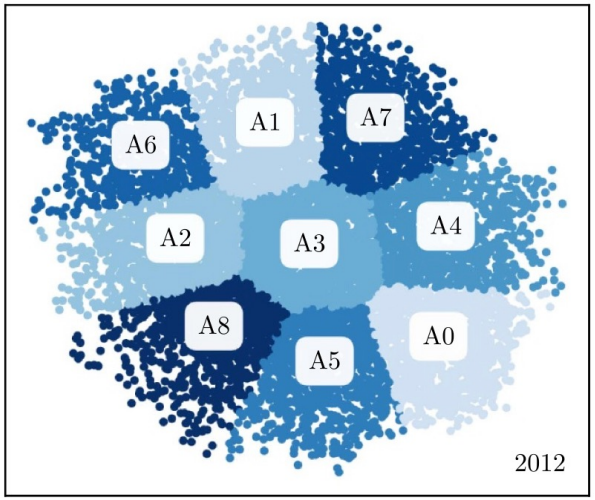


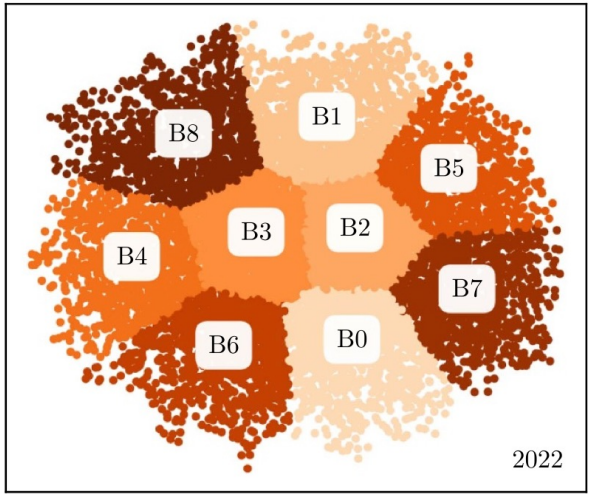
Krenn, M., Buffoni, L., Coutinho, B. et al. Forecasting the future of artificial intelligence with machine learning-based link prediction in an exponentially growing knowledge network.Nat Mach Intell 5, 1326–1335 (2023). https://doi.org/10.1038/s42256-023-00735-0

Discovering emergent connections in quantum physics research via dynamic word embeddings



Felix Frohnert *et al* 2025 *Mach. Learn.: Sci. Technol.* **6** 015029

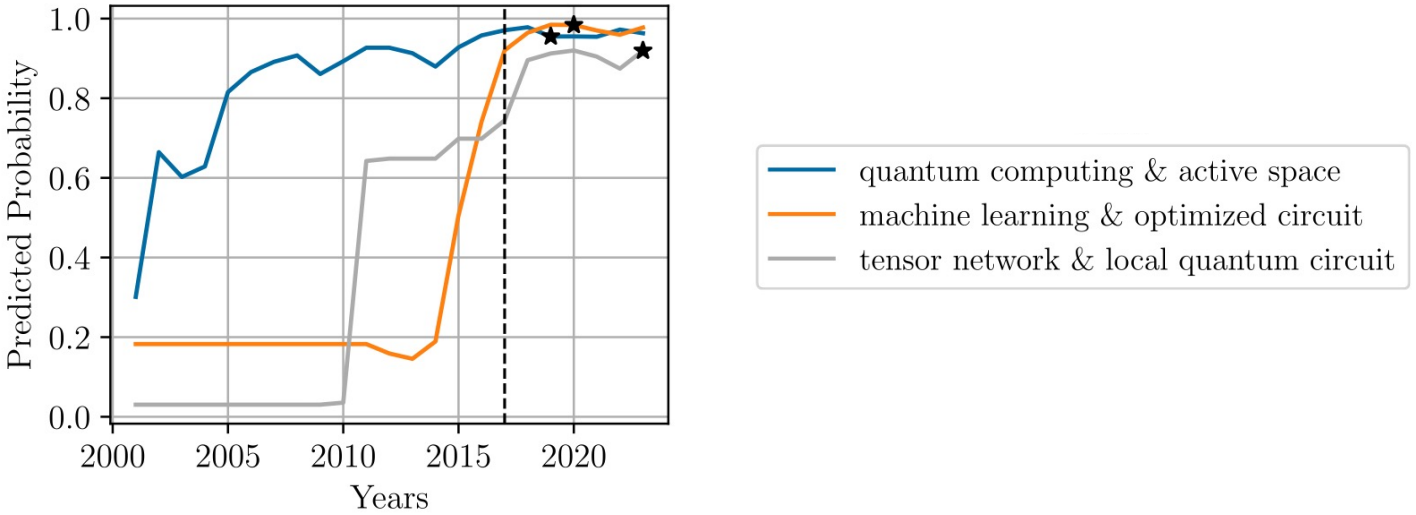




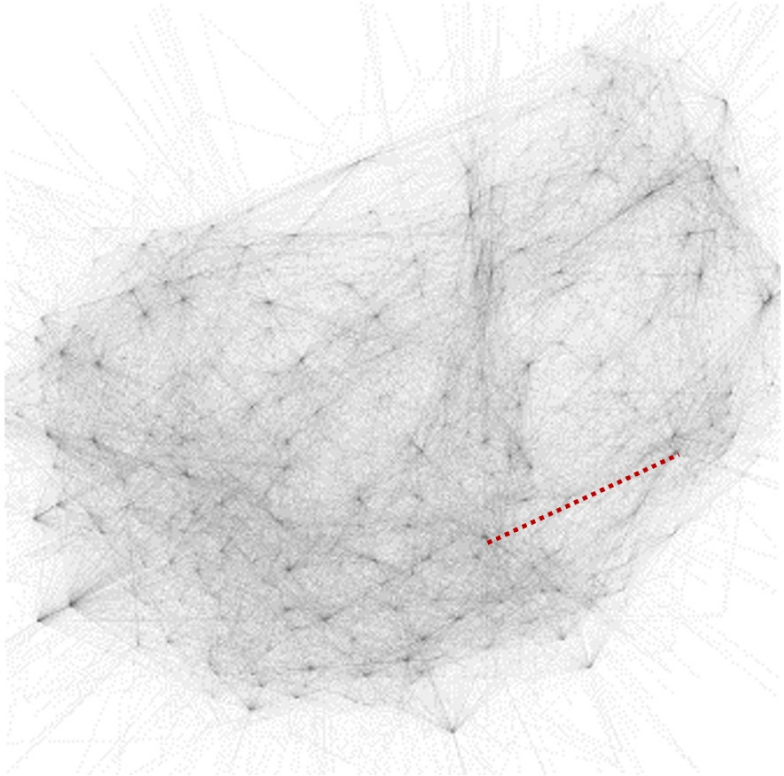
A0	quantum memory	quantum processor	quantum computation
A1	erasure channel	butterfly network	quantum fingerprinting
A2	gallium arsenide	nitrogen atom	quantum dot emission
A3	gaussian wavepacket	quantum propagator	coherent state
A4	phase noise	laser linewidth	cavity resonance
A5	compton scattering	kapitza dirac effect	electron interferometer
A6	ornstein uhlenbeck	white gaussian noise	stochastic dynamic
A7	clifford group	quantum compiler	anyonic model
A8	wavelength multiplexing	laser pumping	quantum repeater

B0	electric field	charged particle	electron wave
B1	ground state wave function	exact diago-nalization	hartree fock theory
B2	quantum computer	quantum speedup	quantum advantage
B3	silicon quantum dot	electron spin resonance	metal oxide semiconductor
B4	photonic integration	satellite com-munication	network capacity
B5	dressed state	rabi oscillation	vacuum splitting
B6	quantum chaotic	prethermal state	ergodic system
B7	quantum logic	boolean algebra	logical connective
B8	laser interferometer	gravitational wave	squeezed quadrature

Model	AUC	Reference	Knowledge graph	Dynamic embedding	Feature type
Word-Dynamic	0.88	Ours	No	Yes	Automatic
Word-Static	0.79	[23]	No	No	Automatic
Word-Hand	0.57	[17]	No	No	Handcrafted
Knowledge-Hand	0.82	[14]	Yes	No	Handcrafted
Knowledge-ProNE	0.79	[15]M7A	Yes	No	Automatic
Knowledge-Node	0.78	[15]M7B	Yes	No	Automatic

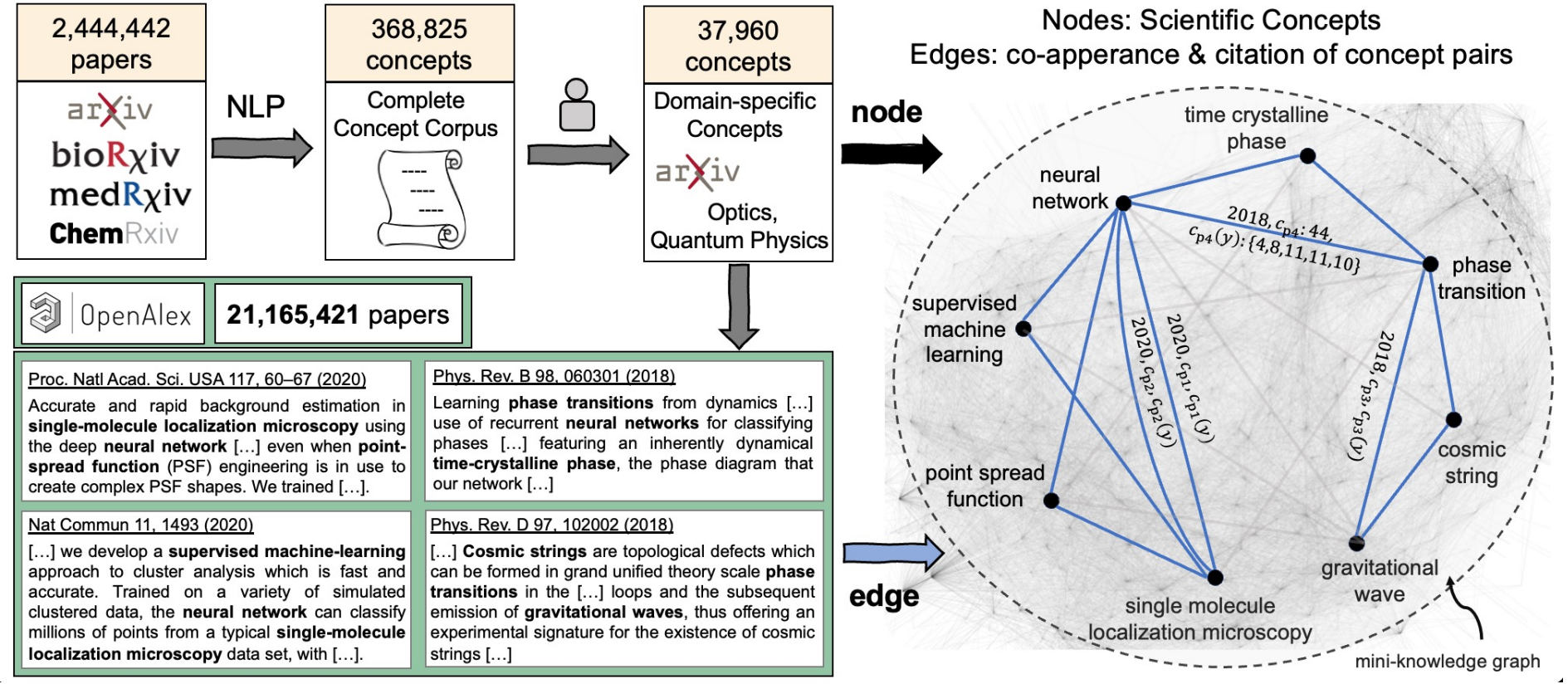


Felix Frohnert *et al* 2025 *Mach. Learn.: Sci. Technol.* **6** 015029

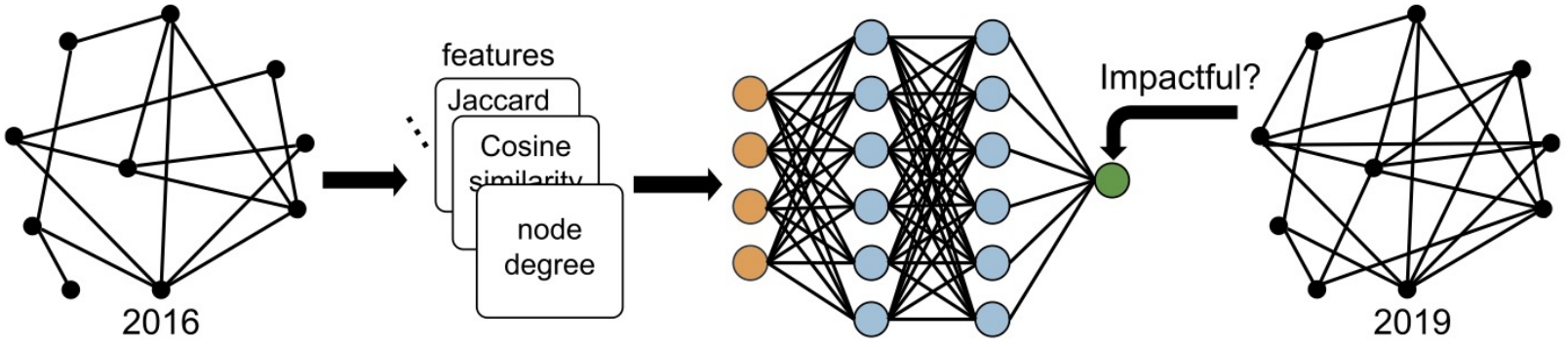


Which scientific concepts, that have never been investigated jointly, will lead to the most impactful research?

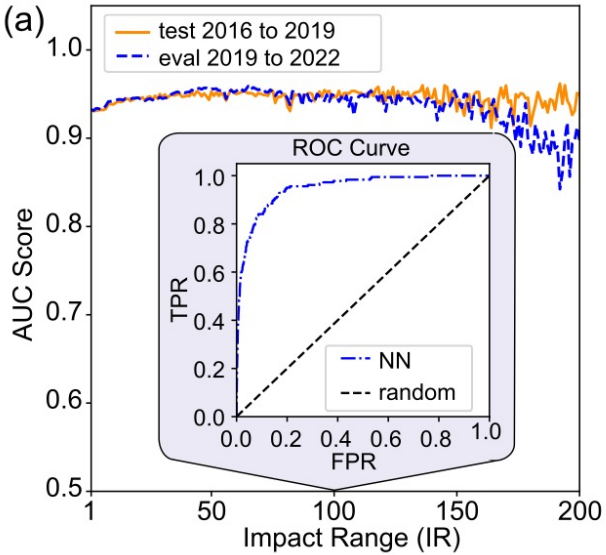
How interesting are these AI-generated ideas for researchers, and how can we improve their quality?

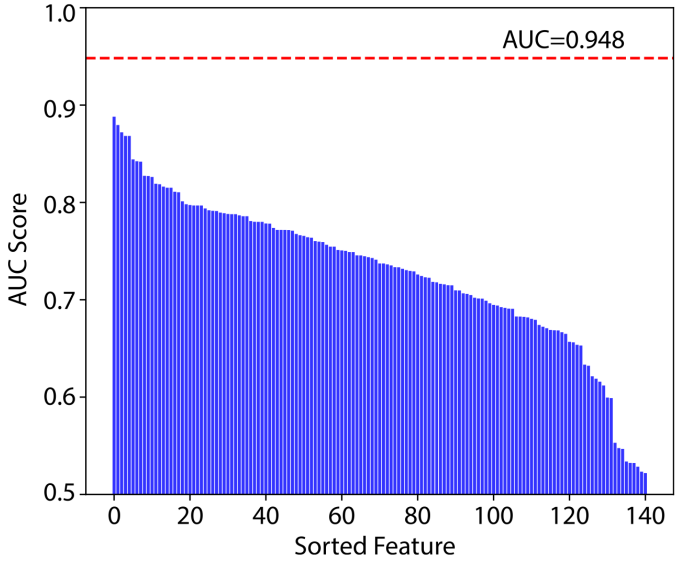


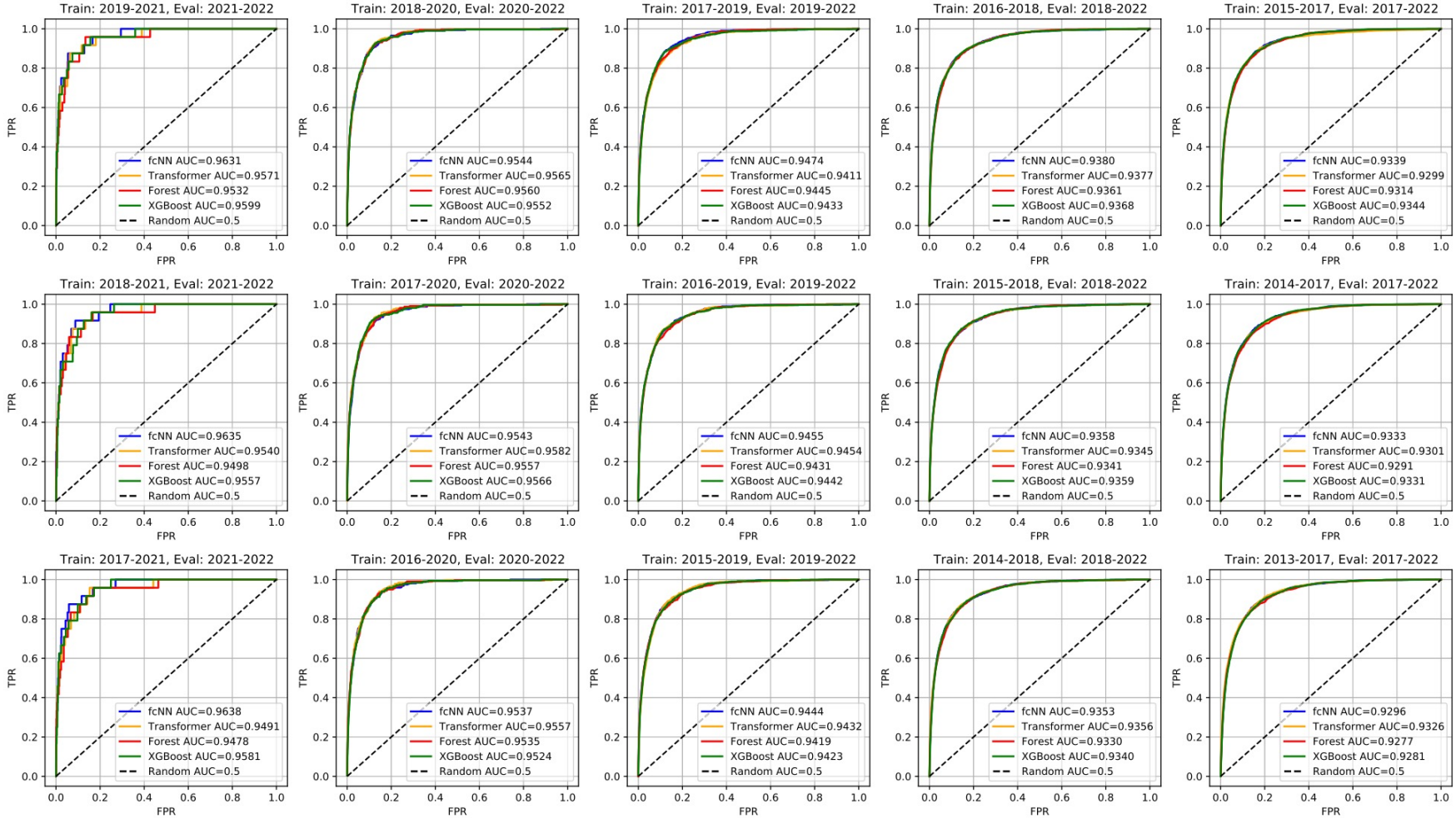
Gu, Krenn, Forecasting high-impact research topics via machine learning on evolving knowledge graphs,
Machine Learning: Science & Technology, 6, 025041 (2025)



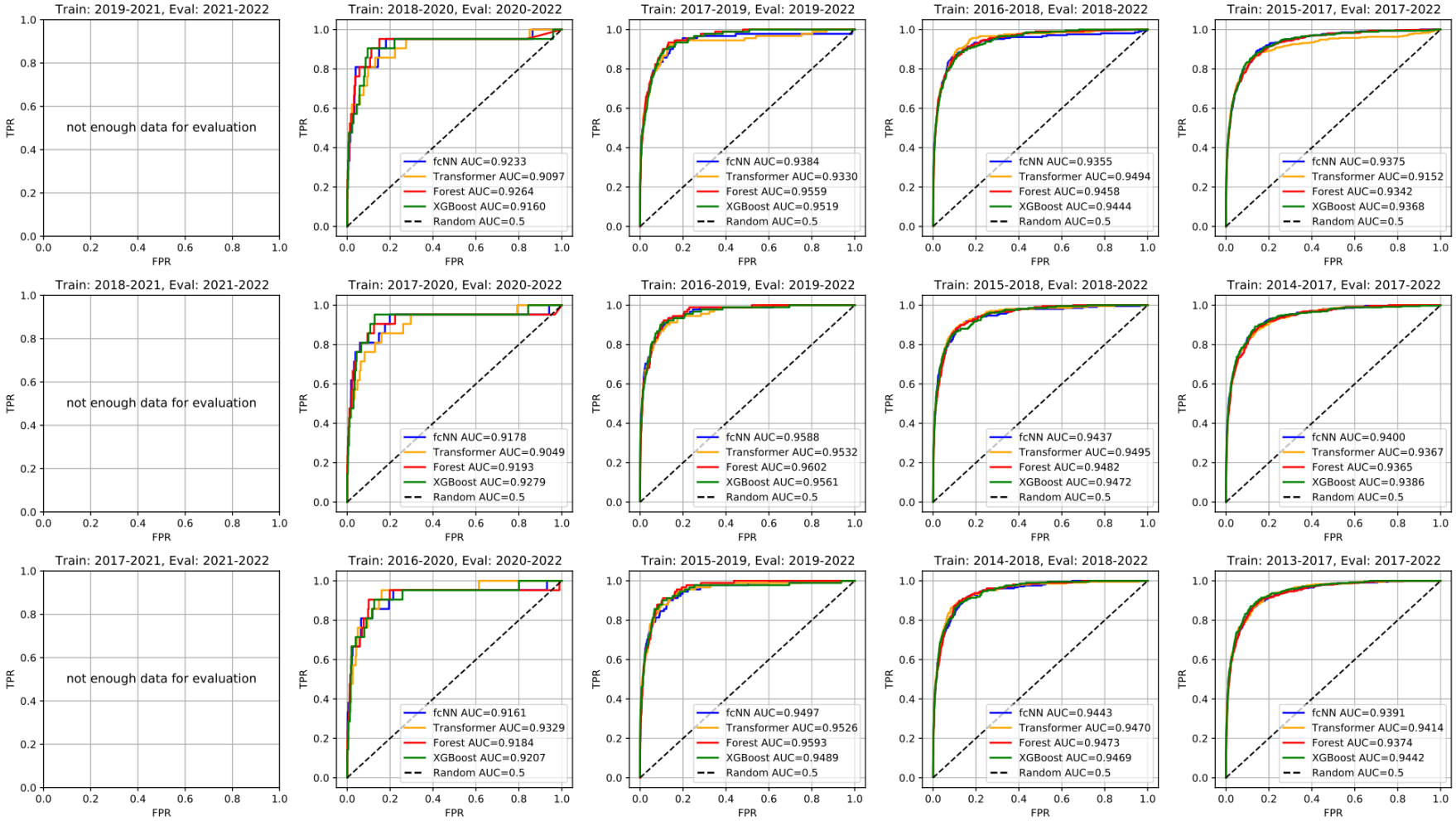
train dataset: 2016 -- 2019; test dataset: holdout data 2016 -- 2019; eval dataset: 2019 --2022





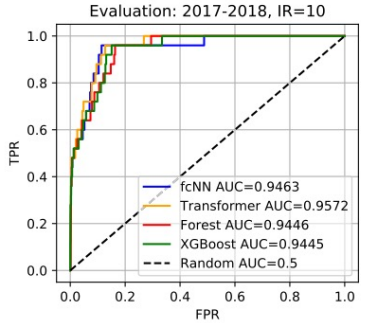


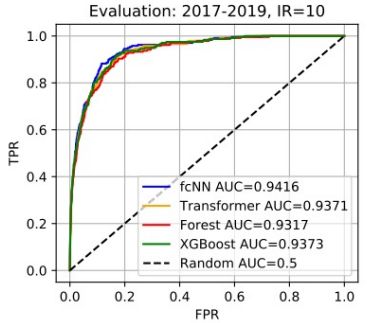
Impact range IR=10

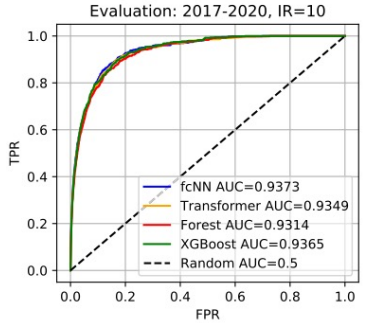


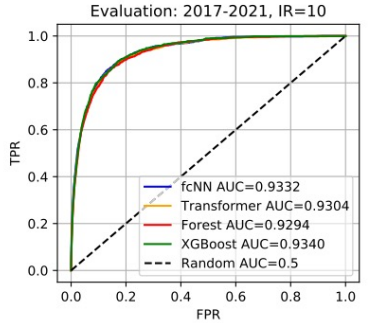
Impact range IR=50

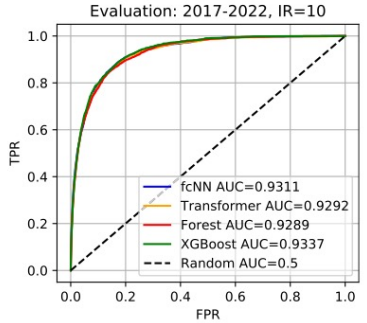
Training from 2014 -> 2017

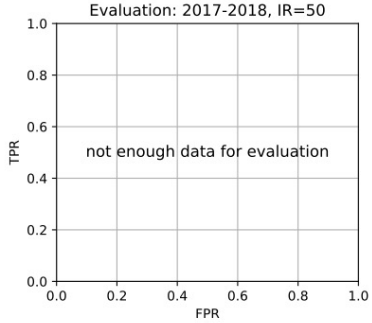


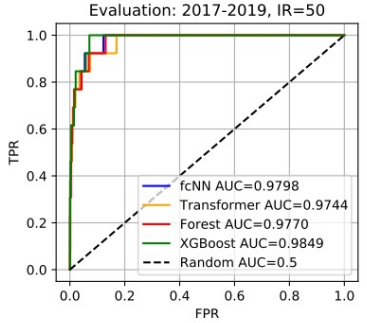


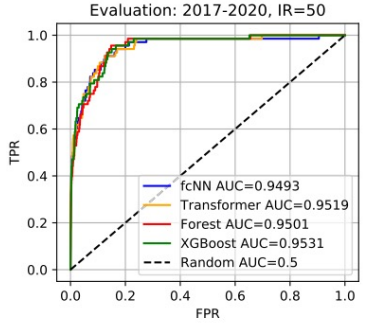


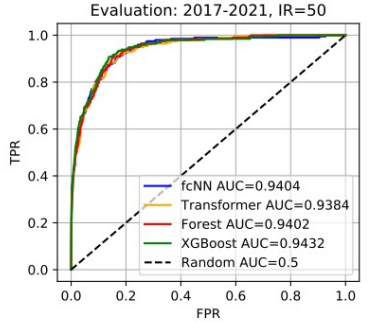


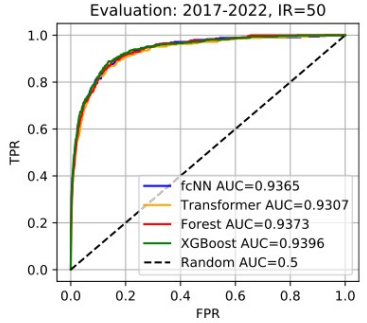


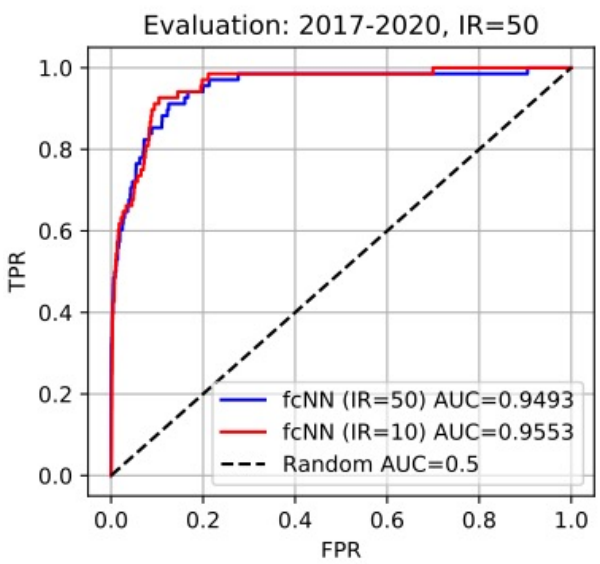


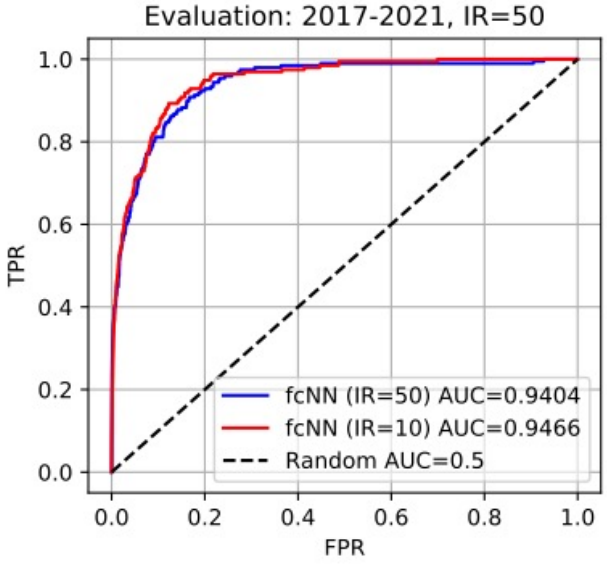


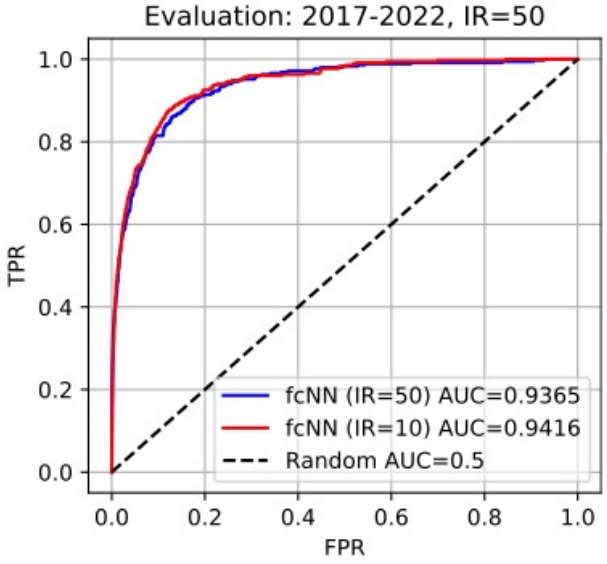


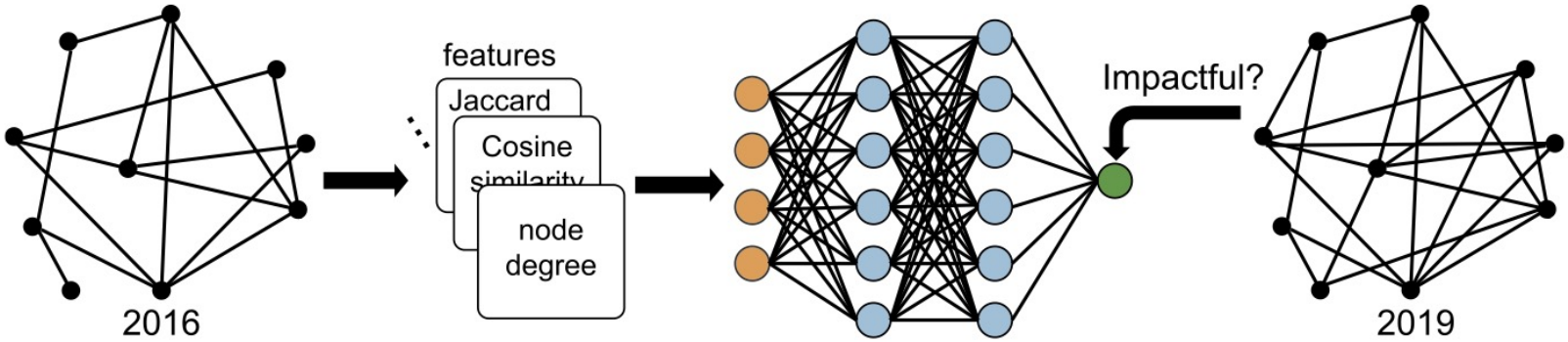





Evaluation: 2017-2020, IR=50
TPR
FPR
fcNN (IR=50) AUC=0.9493
fcNN (IR=10) AUC=0.9553
Random AUC=0.5


Evaluation: 2017-2021, IR=50
TPR
FPR
fcNN (IR=50) AUC=0.9404
fcNN (IR=10) AUC=0.9466
Random AUC=0.5


Evaluation: 2017-2022, IR=50
TPR
FPR
fcNN (IR=50) AUC=0.9365
fcNN (IR=10) AUC=0.9416
Random AUC=0.5



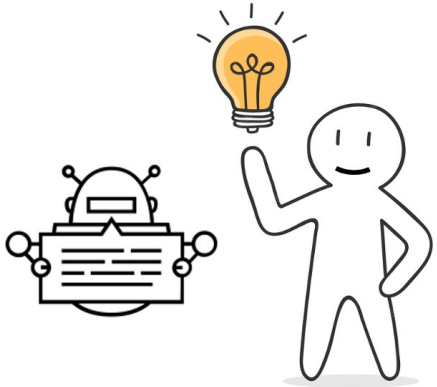
train dataset: 2016 -- 2019; test dataset: holdout data 2016 -- 2019; eval dataset: 2019 --2022

Then, train 2019 for 2022, apply 2022 to 2025!

Suggesting impactful research directions

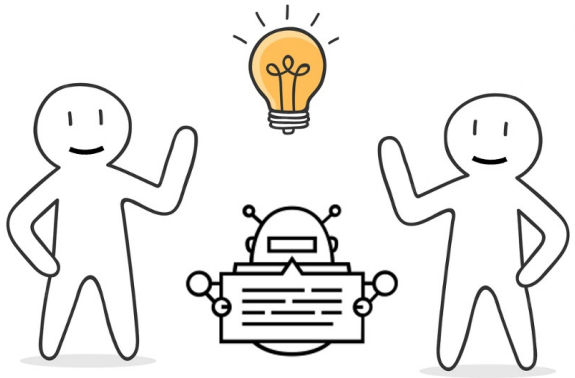
personalized research

At least one concept in your area of interest (used in own paper)

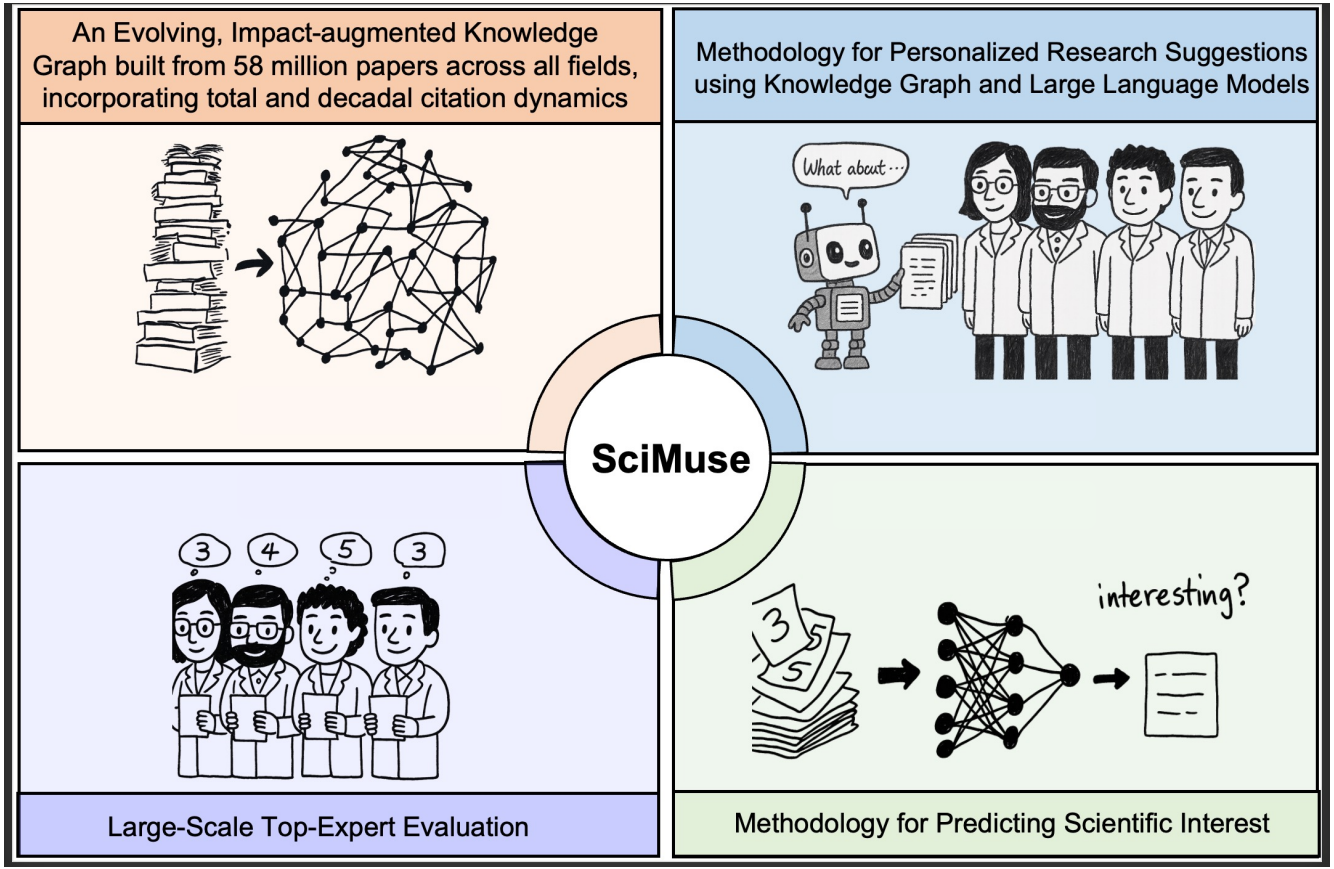


research collaborations

Concept 1 from ResearcherA
Concept 2 from ResearcherB

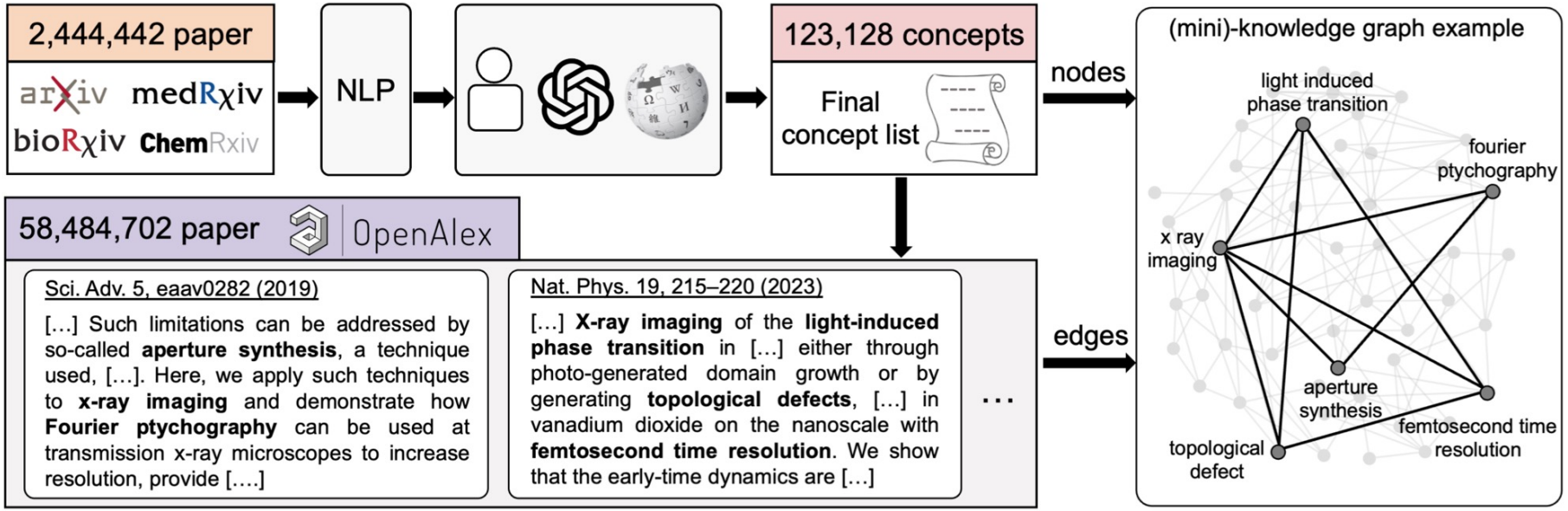


Real Project Ideas? Multidisciplinary Ideas?



Gu, Krenn, *Interesting Scientific Idea Generation Using Knowledge Graphs and LLMs: Evaluations with 100 Research Group Leaders*, arXiv:2405.17044 (2024).

Real Project Ideas? Multidisciplinary Ideas?



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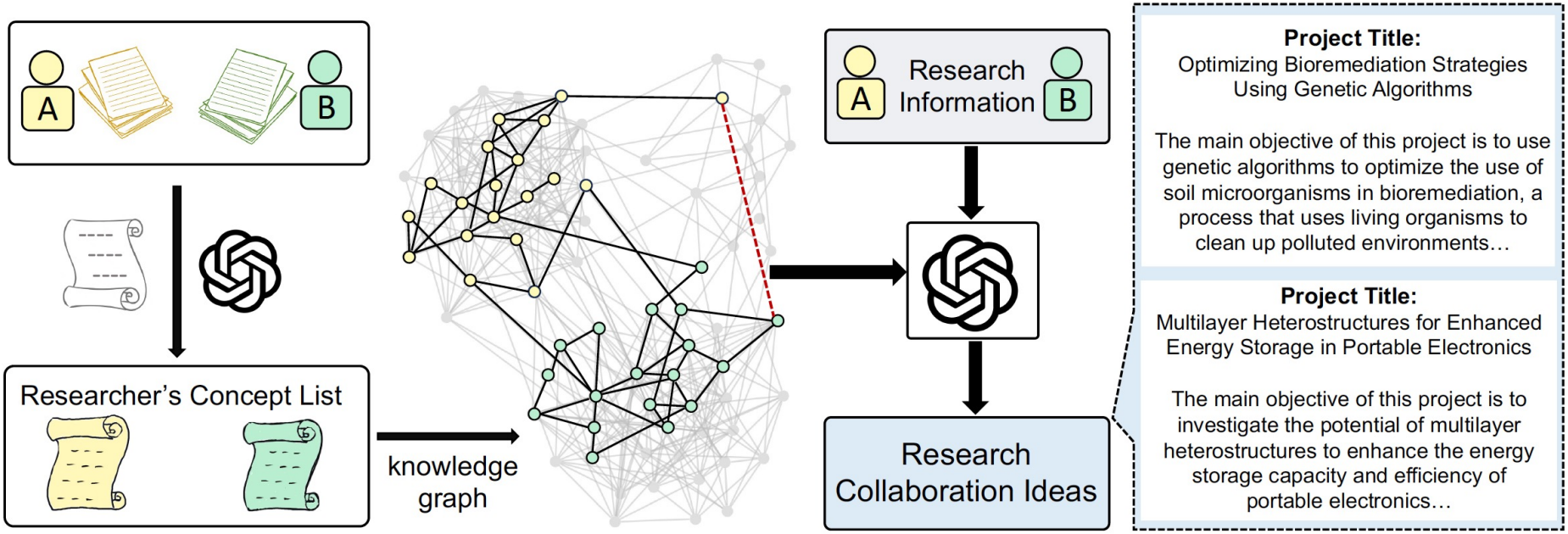
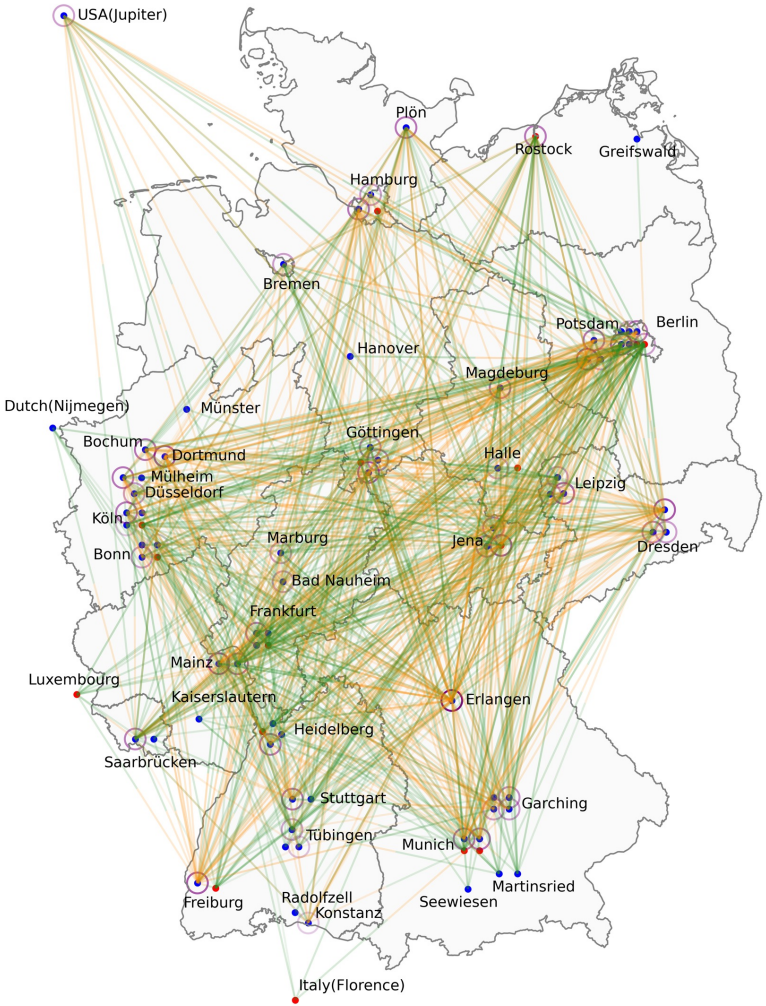
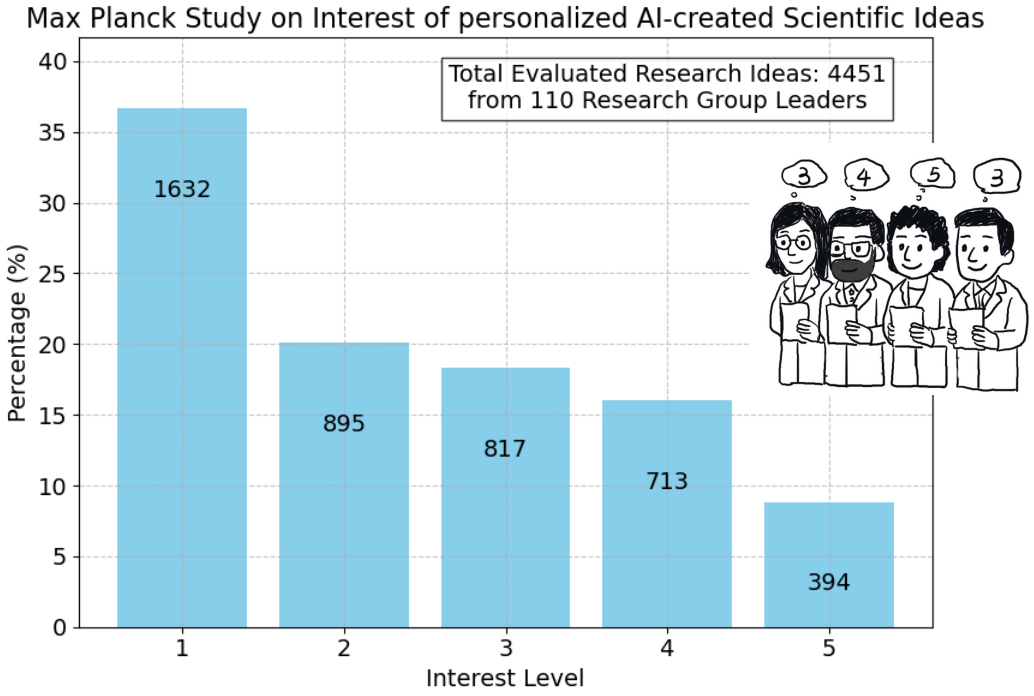
Real Project Ideas? Multidisciplinary Ideas?

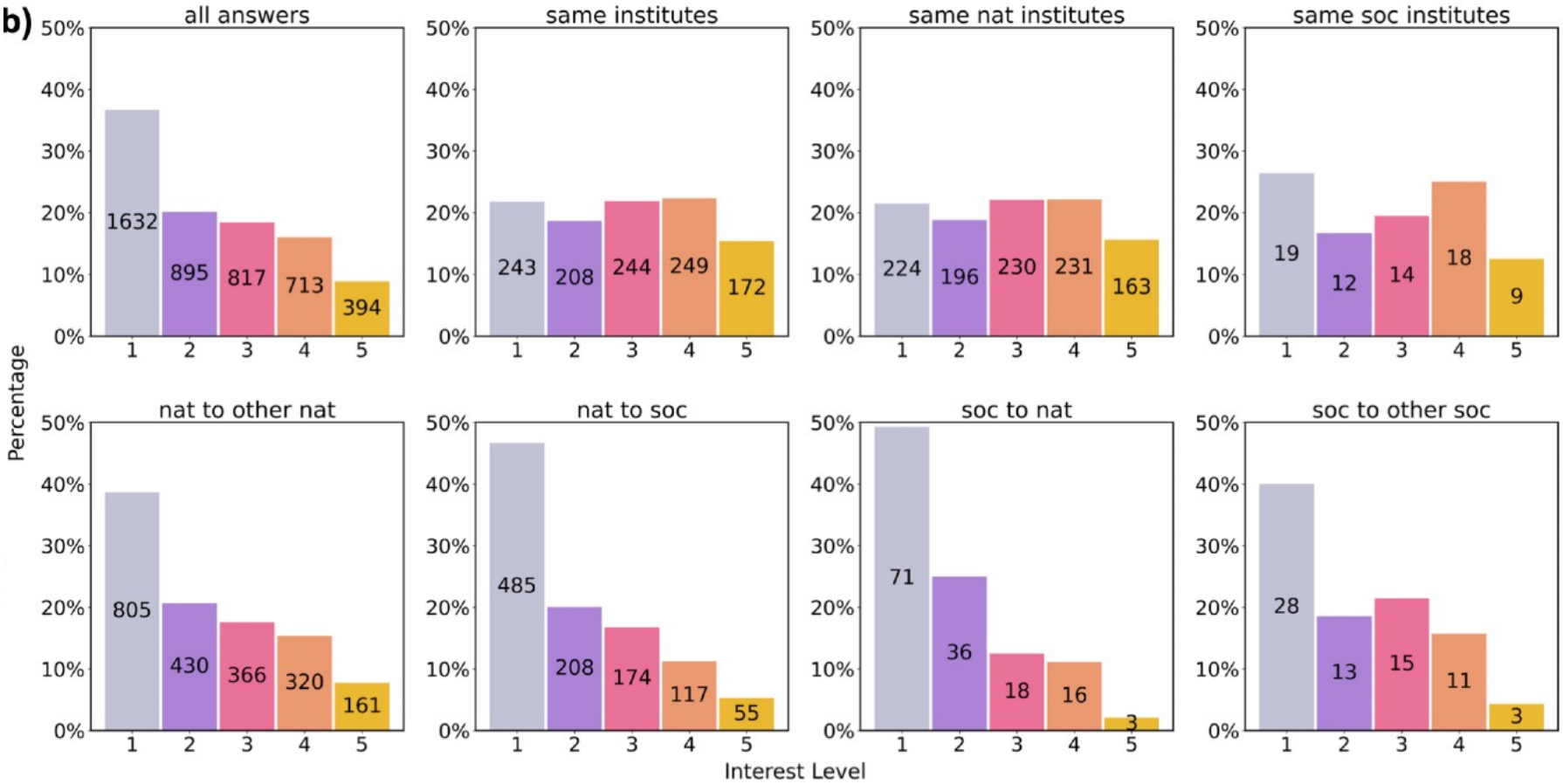


Gu, Krenn, *Interesting Scientific Idea Generation Using Knowledge Graphs and LLMs: Evaluations with 100 Research Group Leaders*, arXiv:2405.17044 (2024).



Human evaluation




b)
all answers
same institutes
same nat institutes
same soc institutes
nat to other nat
nat to soc
soc to nat
soc to other soc
Percentage
Interest Level
0%
10%
20%
30%
40%
50%
1
2
3
4
5
1632
895
817
713
394
243
208
244
249
172
224
196
230
231
163
19
12
14
18
9
805
430
366
320
161
485
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174
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71
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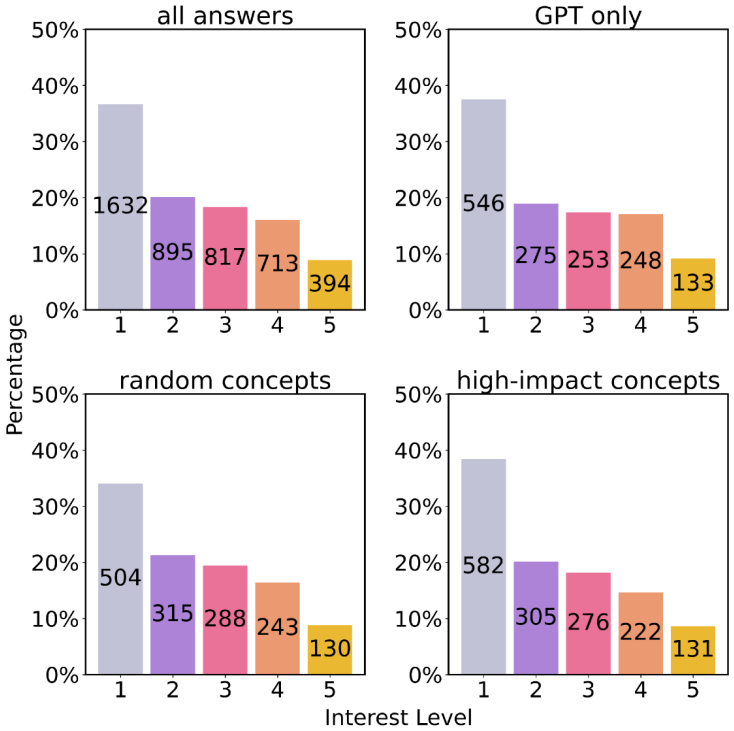
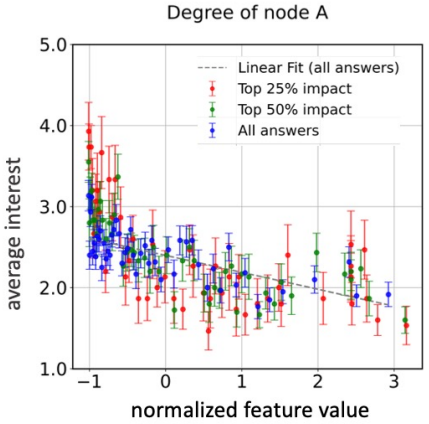
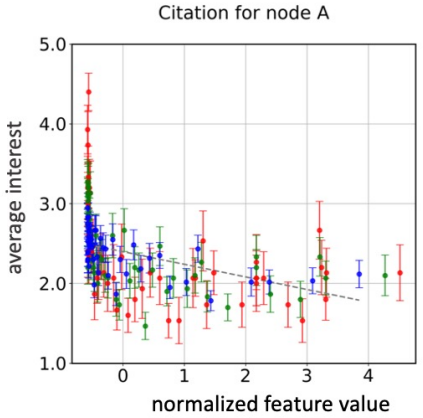
11
3

2,996 suggested research projects, which were created using concept pairs from the knowledge graph







the more connected a concept is within the knowledge graph, the less appealing the research project appears

the more frequently a concept has been cited in the past, the less interesting the project is evaluated

Predictin

super

unsupe

Prompt for Zero-Shot Ranking of Research Ideas

I will present two research ideas. The first idea is for Researchers A1 and B1, and the second idea is for Researchers A2 and B2.
Researchers A1 and A2 will evaluate how interesting they find the respective ideas.
You will determine which of the two suggestions will be considered more interesting.

The suggestions are randomly ordered, and you should evaluate each suggestion independently and without bias.

Researcher A1 Context and Suggestion 1:
Here are a few papers of Researcher A1:
(papersA1)

Suggestion 1: [suggestion1]

Summary for Researcher A1: Provide a one-sentence summary of Suggestion 1 in the context of Researcher A1.

Researcher A2 Context and Suggestion 2:
Here are a few papers of Researcher A2:
(papersA2)

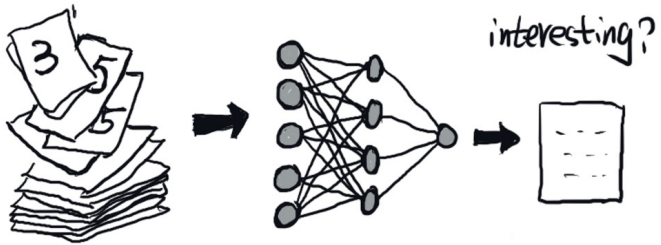
Suggestion 2: [suggestion2]

Summary for Researcher A2: Provide a one-sentence summary of Suggestion 2 in the context of Researcher A2.

Evaluation:
Based on the summaries and the research interests of A1 and A2, evaluate which suggestion is more likely to be ranked higher in terms of interest.
Result: If Suggestion 1 is ranked higher by Researcher A1 than Suggestion 2 is by Researcher A2, write 'RESULT: SUGGESTION 1'. Otherwise, write 'RESULT: SUGGESTION 2'.
Remember, the suggestions are randomly ordered, and your evaluation should be impartial and based solely on the research interests of A1 and A2.

Predicting Researcher's Scientific Interest

supervised neural networks trained on human evaluations

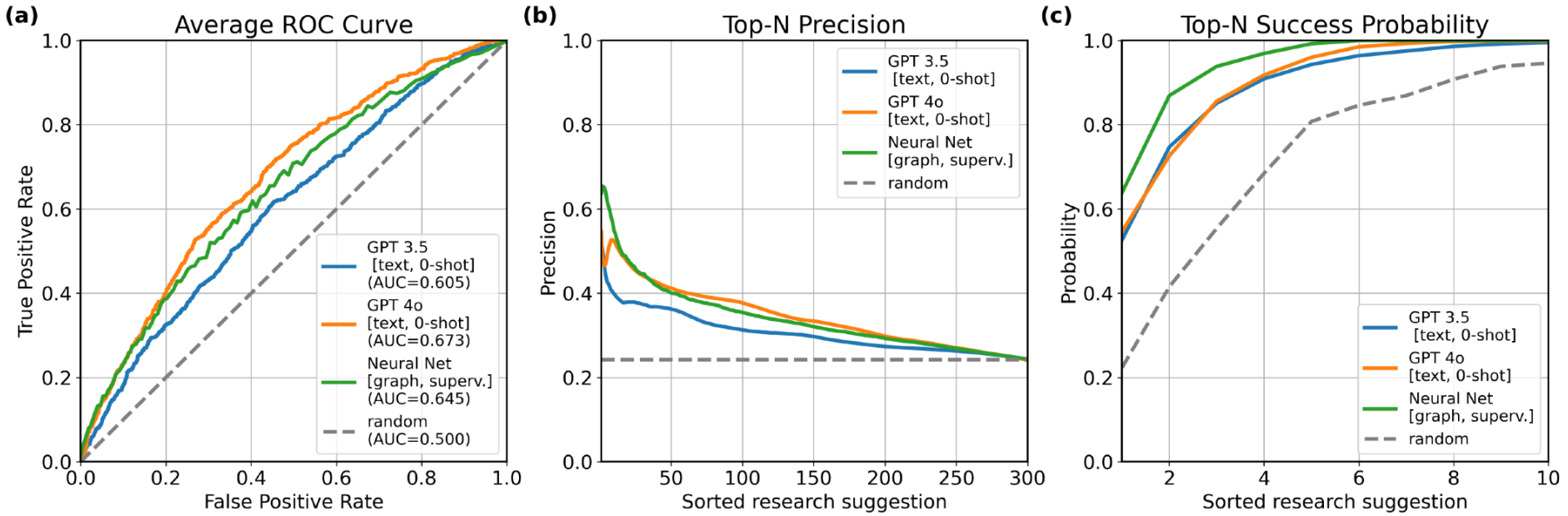


unsupervised zero-shot ranking with large-language models

Research ideas tournament/ELO ranking

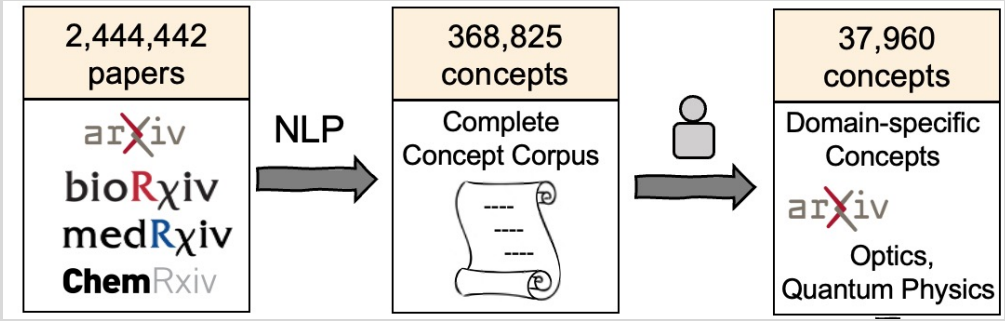
random pair from the 2996 ideas

Which suggestion is more likely to be ranked higher in terms of interest of the researchers?

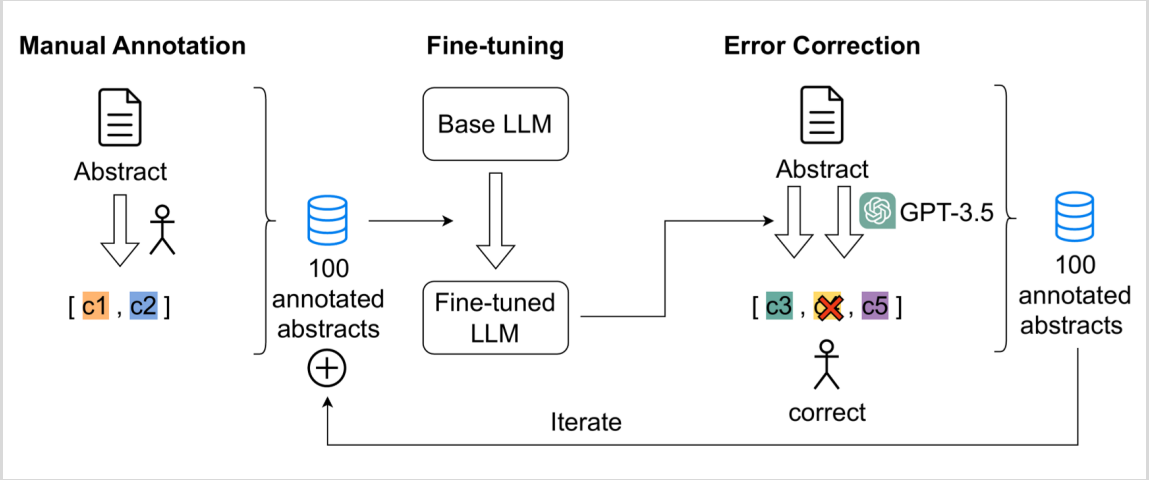

(a)
Average ROC Curve
True Positive Rate
False Positive Rate
GPT 3.5 [text, 0-shot] (AUC=0.605)
GPT 4o [text, 0-shot] (AUC=0.673)
Neural Net [graph, superv.] (AUC=0.645)
random (AUC=0.500)
(b)
Top-N Precision
Precision
Sorted research suggestion
GPT 3.5 [text, 0-shot]
GPT 4o [text, 0-shot]
Neural Net [graph, superv.]
random
(c)
Top-N Success Probability
Probability
Sorted research suggestion
GPT 3.5 [text, 0-shot]
GPT 4o [text, 0-shot]
Neural Net [graph, superv.]
random

Concept extraction

Rapid Automatic Keyword Extraction (RAKE)

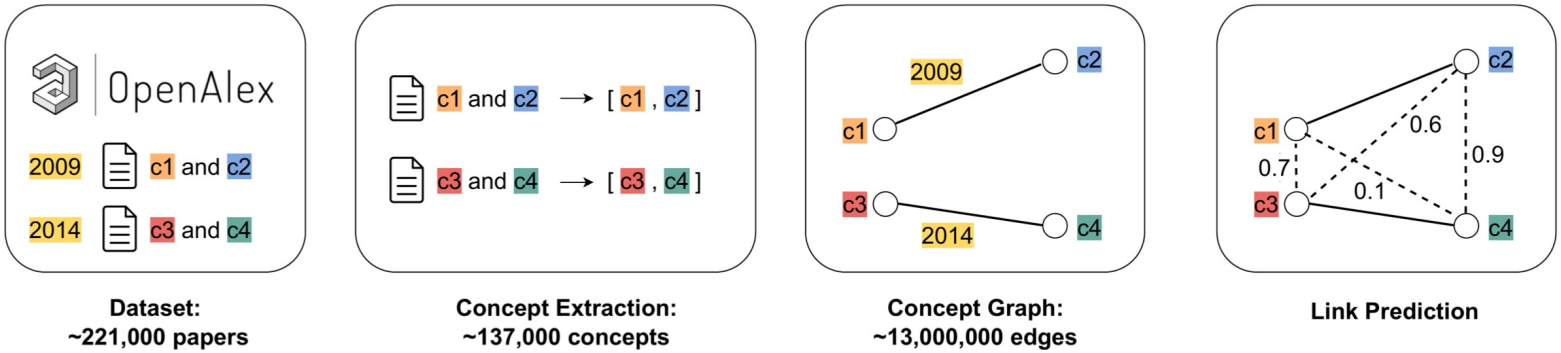


Fine-tuned LLMs



Marwitz, Thomas, et al. "Predicting New Research Directions in Materials Science using Large Language Models and Concept Graphs." *arXiv:2506.16824* (2025).

Predicting New Research Directions in Materials Science

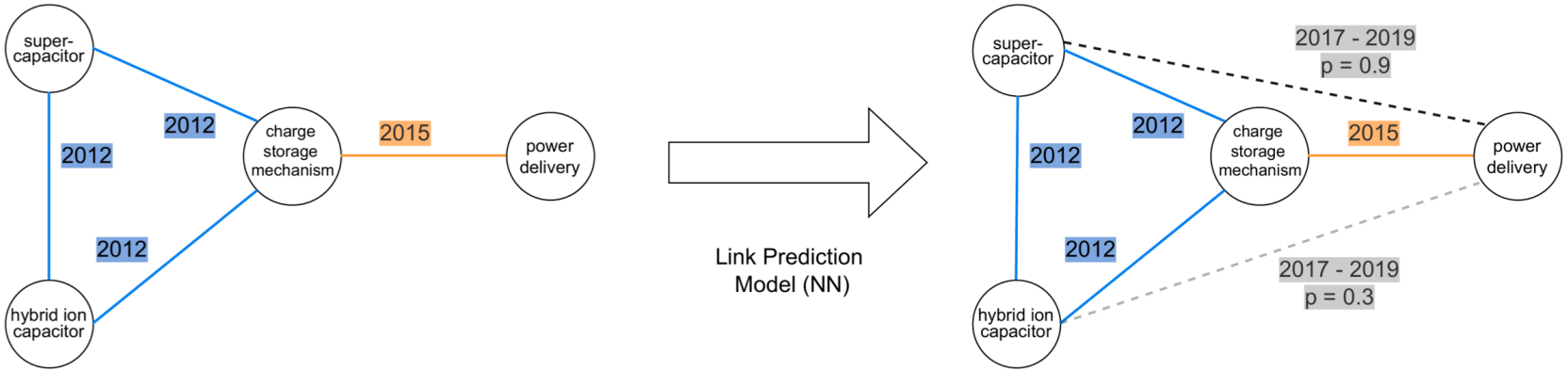


Dataset:
~221,000 papers

Concept Extraction:
~137,000 concepts

Concept Graph:
~13,000,000 edges

Link Prediction



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http://inspire.aimat.science

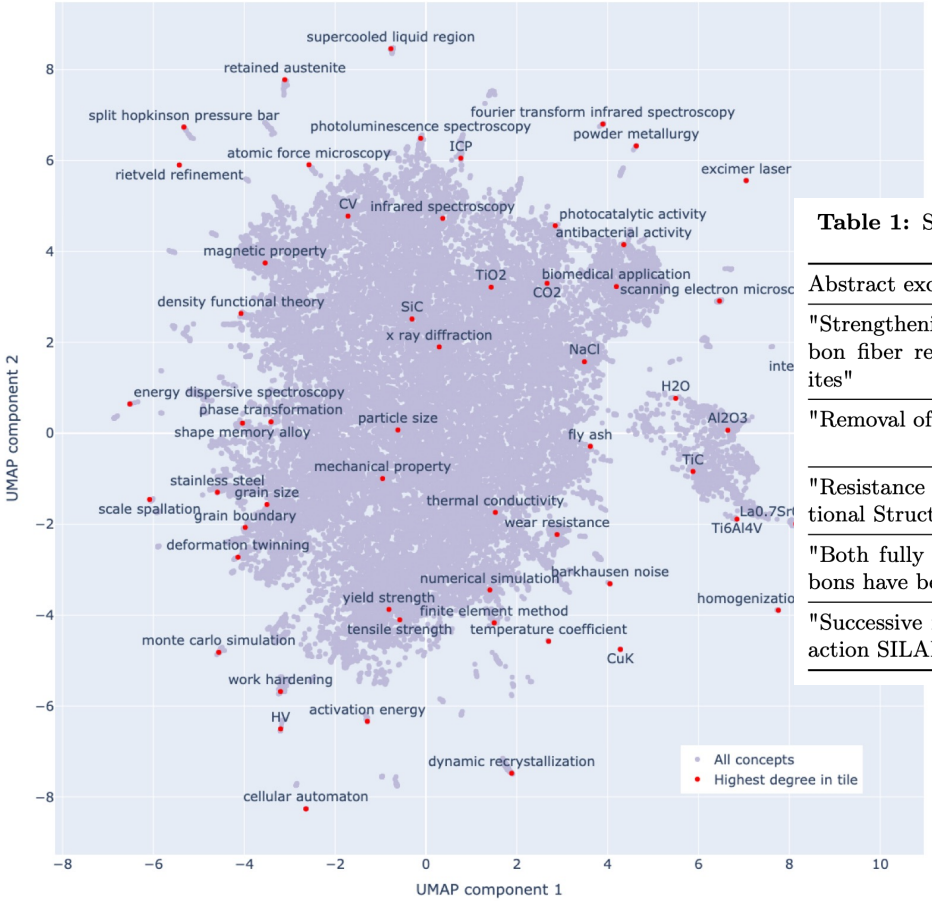
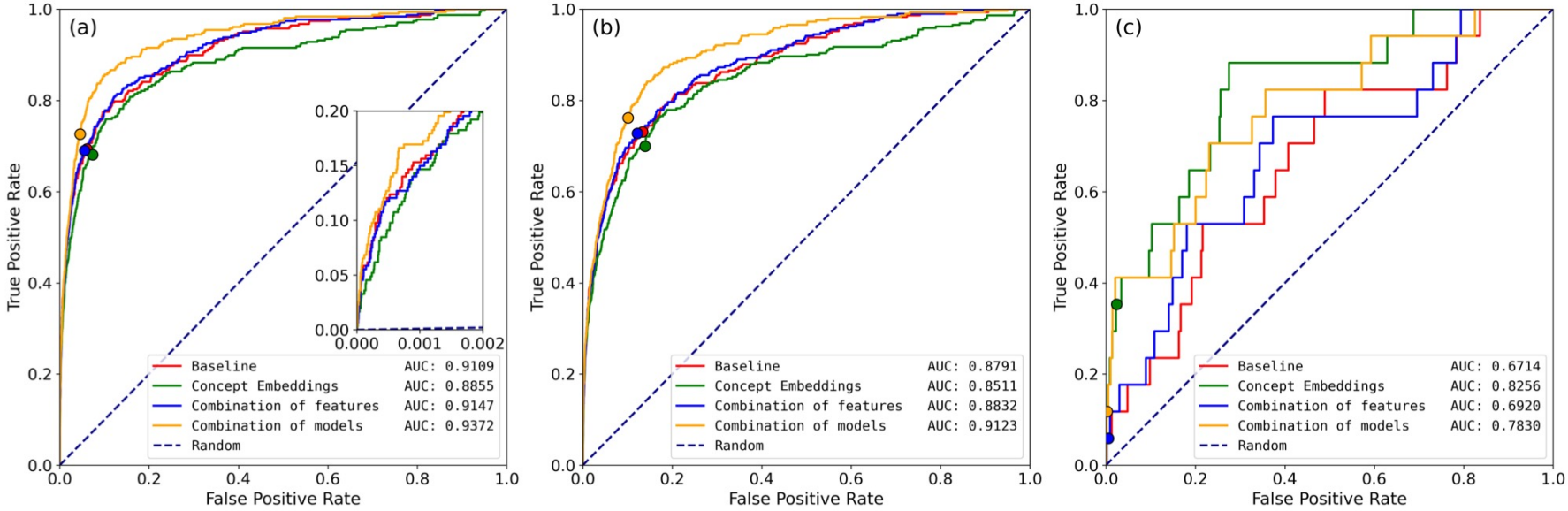


Table 1: Selected examples of abstracts and concepts extracted by our fine-tuned Llama-2-13B model.

Abstract excerpt	Extracted concept	Note
"Strengthening mechanisms in short carbon fiber reinforced Nb/Nb5Si3 composites"	carbon fiber reinforcement	Nominalisation
"Removal of Metal Impurities from the"	metal impurity removal	Removal of 'of', singular normalization
"Resistance of Al2O3Coatings on Functional Structure"	al2o3 coating	Singular normalization, dealing with wrong formatting
"Both fully and partially amorphous ribbons have been obtained"	fully amorphous ribbon, partially amorphous ribbon	Normalization of 'and' in concepts
"Successive ionic layer adsorption and reaction SILAR trend for"	successive ionic layer adsorption and reaction	Extraction of long form without abbreviation in same concept



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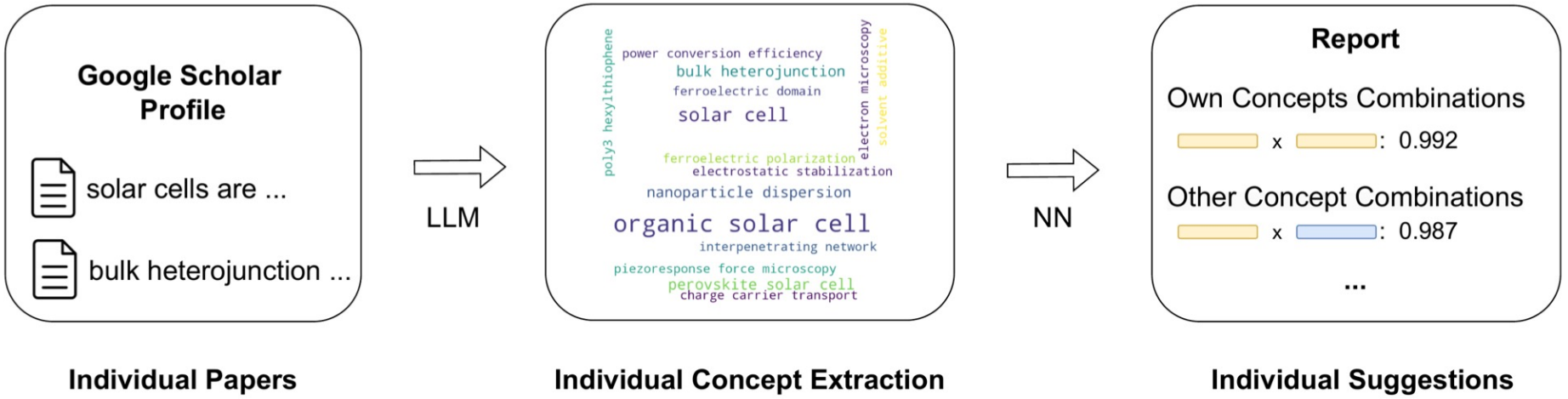
Idea Generation



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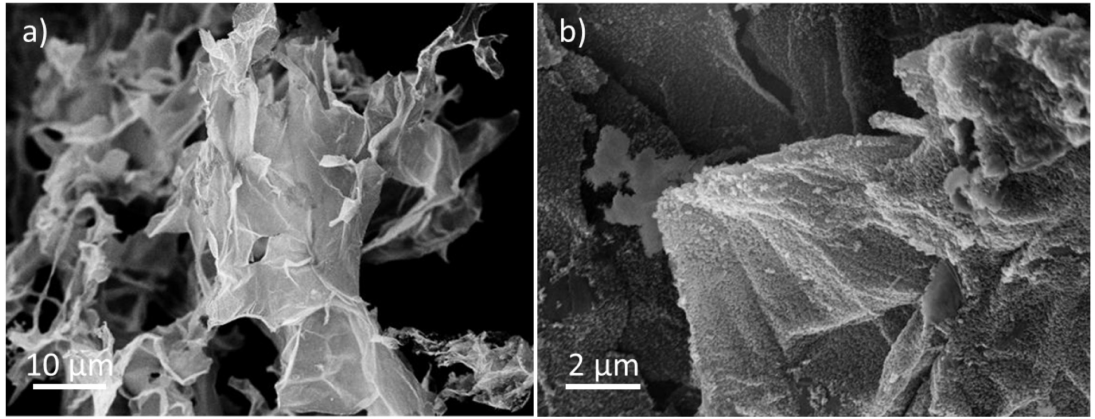
Human Evaluation

A1 (known), A2 (trivial), B (nonsense), and C (interesting)

Section	Category	Occurance	
		Number	Fraction
$S_{\text{own}\times\text{own}}$	A1	31	30.7%
	A2	21	20.8%
	B	25	24.8%
	C	24	23.7%
$S_{\text{own}\times\text{other}}$	A1	17	22.1%
	A2	8	10.4%
	B	35	45.4%
	C	17	22.1%
$S_{(\text{many own})\times\text{other}}$	A1	6	23.1%
	A2	1	3.9%
	B	3	11.5%
	C	16	61.5%
$S^{\text{filtered}}_{\text{own}\times\text{other}}$	A1	17	19.3%
	A2	6	6.8%
	B	36	40.9%
	C	29	33.0%

Suggestion: "Conventional ceramic" + "Graphene oxide"

This suggestion associates "conventional ceramics" with "graphene oxide", two domains seldom combined. Conventional oxide ceramics provide chemical, thermal, and structural stability. Graphene oxide offers a high-surface-area, electronically conductive carbon framework. Their union could yield composites marrying ceramic robustness with rapid charge and heat transport, relevant to batteries, catalysis, and thermal barriers. Existing studies mix pre-synthesised oxides with graphene derivatives, giving limited interfacial contact. Our unpublished autoclave route aligns with the AI hint: $Fe(CO)_5$ decomposes on graphene oxide, forming a ≈200 nm iron-oxide shell on multilayer graphene (see Figure 8. The in-situ process creates intimate oxide–graphene interfaces and a continuous conductive network. Electrochemical tests show high reversible capacity and enhanced redox kinetics in Li-ion conversion cells. These findings demonstrate the model's capacity to provide inspiration for overlooked but feasible synthesis strategies. Systematic exploration of AI-highlighted ceramic/graphene hybrids may accelerate multifunctional material discovery.



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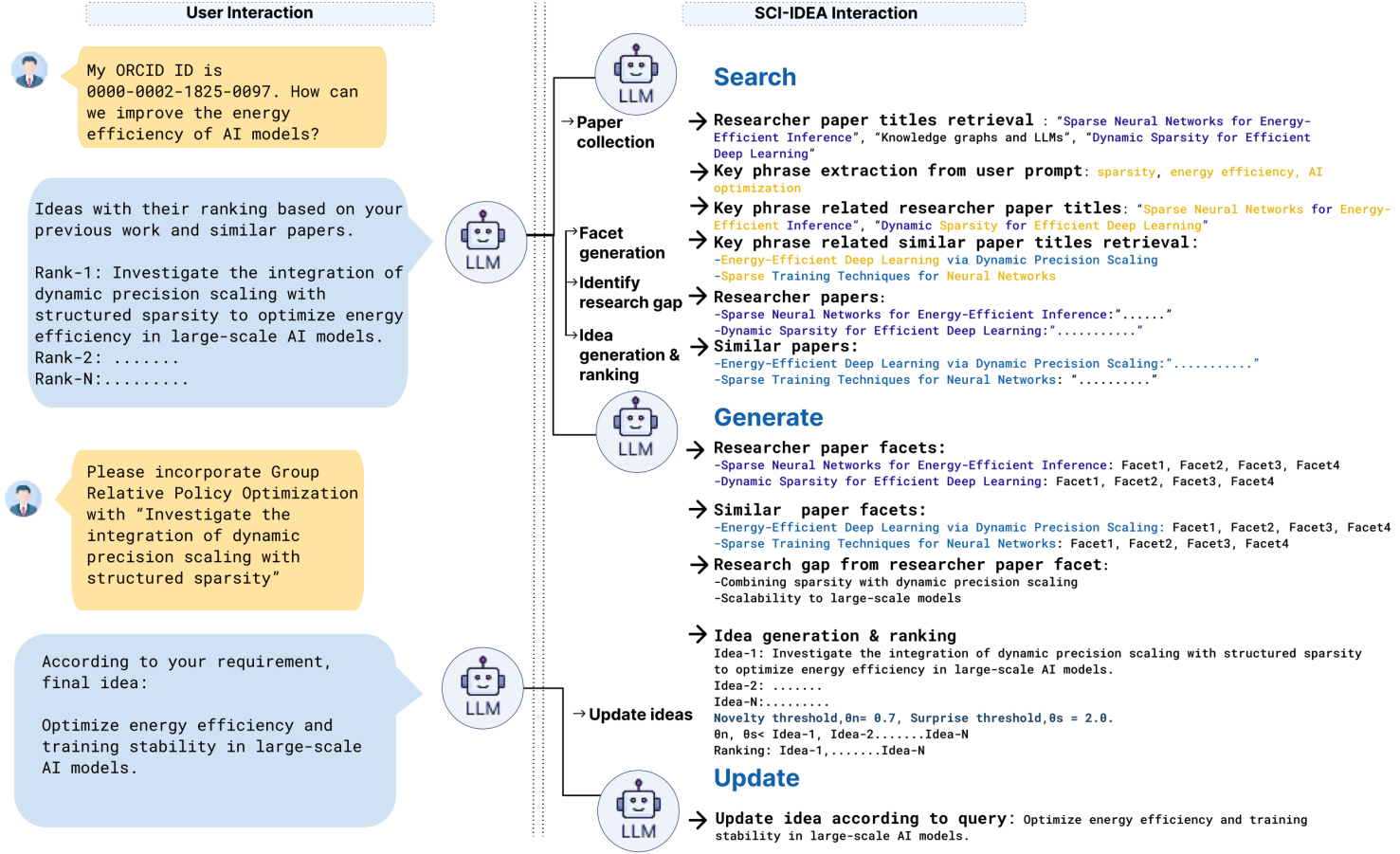

Farhana Keya, et al. "SCI-IDEA: Context-Aware Scientific Ideation Using Token and Sentence Embeddings." *arXiv:2503.19257* (2025).

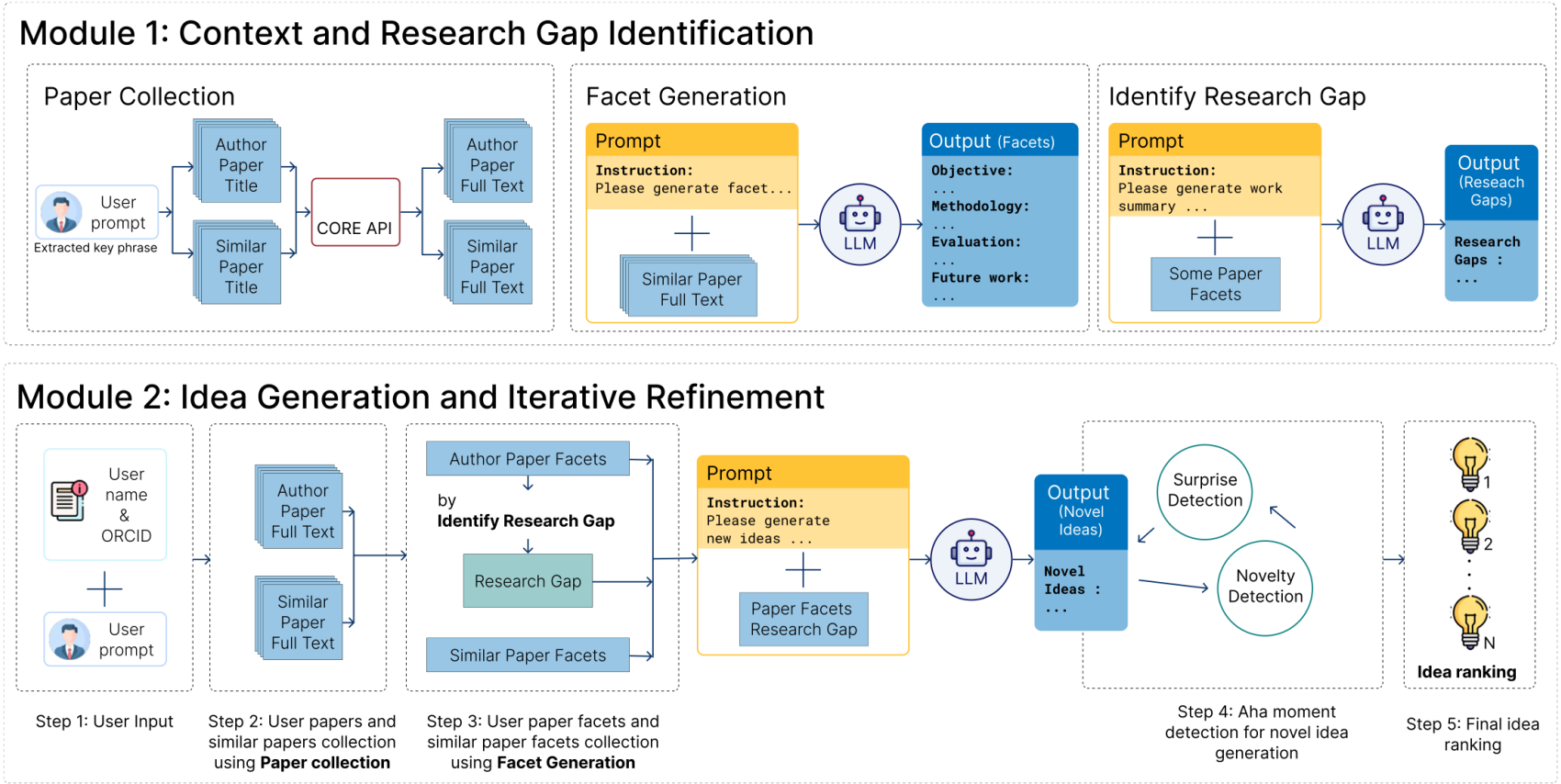

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Table 4: Scientific LLM systems and agents. We identify the studies we classified as an agent with the icon [419] and multi-agent systems with the icon . Ⓦ, Ⓢ, and Ⓛ[420] mean the agent bases his behavior on sensory, short, and long memory components, respectively. Besides the textual capabilities of LLM-based agents, [421] and [422] mean the agent has additional audio and visual perception, respectively.

Agent	Memory	Planning	Reasoning	Action	Release
LLaMP[423]	ⓈⓁ		✓	Tools for database access, literature search, and atomistic simulations	2024.06
CRISPR-GPT[424]	Ⓢ	✓	✓	Tool for gene editing experiments design	2024.04
TAIS[425]	Ⓢ	✓	✓	Tools for gene expression data analysis	2024.02
SciAgent[426]	ⓌⓁ	✓	✓	Trained Mistral for tool usage. Evaluated it using MathToolBench's tools	2024.02
STORM[427]	ⓈⓁ	✓	✓	Article writing using retrieval from multi-LLM conversations and pre-generated outline	2024.02
Völker et al.[428]	Ⓢ			Regression with ICL and text retrieval	2024.02
ProtAgent[429]	ⓈⓁ	✓	✓	Tools for proteins information retrieval, analyzing, *de novo* design, and 3D folded structure generation	2024.01
PaperQA[430]	ⓈⓁ	✓	✓	Tools to search the scientific literature, gather evidence, and answer questions	2023.12
WikiCrow[431]	ⓈⓁ	✓	✓	Uses PaperQA as a tool	2023.12
Coscientist[46]	Ⓢ	✓	✓	Tools for running Python code, web-searching, and interacting with lab equipment	2023.12
Eunomia[432]	Ⓢ		✓	Tools for literature and dataset searching and a chain-of–verification loop	2023.12
CALMS[433]	Ⓛ			Tools for using the Materials Project API, designing experiments, and using a hardware API to perform the experiment	2023.12

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STORM [427]	ⓈⓁ	✓	✓		
Völker et al. [428]	Ⓢ				
ProtAgent [429]	ⓈⓁ	✓	✓		
PaperQA [430]	ⓈⓁ	✓	✓		
WikiCrow [431]	ⓈⓁ	✓	✓		
Coscientist [46]	Ⓢ	✓	✓		
Eunomia [432]	Ⓢ		✓		
CALMS [433]	Ⓛ				

CoQuest [434]	Ⓢ		✓	Research question generations and tools for literature visualization using a graph organization	2023.10
eXpertAI [286]	Ⓢ		✓	Tools for applying XAI methods	2023.11
BioPlanner [435]	Ⓢ	✓	✓	Tool for protocol searching in the BioProt dataset	2023.10
IBM ChatChem [436]	Ⓢ			Tools for cheminformatics and accessing GT4SD and HuggingFace models	2023.09
ChatMOF [437]	ⓌⓈ		✓	Tools for database search, property prediction, and MOF's structure generation	2023.08
AmadeusGPT [45]	ⓈⓁ		✓	Tools for writing and executing code for computer vision, machine learning, and spatial-temporal reasoning	2023.07
i-Digest [438]	Ⓢ			Uses the whisper model to process audio transcription from classes and write summaries and following up questions	2023.06
BOLLaMa [439]	Ⓦ			Implements an LLM interface to ease the usage of their BO code	2023.06
text2concrete [440]	Ⓢ			Uses ICL to predict compressive strength from concrete formulation	2023.06
MAPI_LLM [441]	ⓈⓁ		✓	Database access and LLM prediction using ICL	2023.06
BO-LIFT [442]	Ⓢ			Regression using ICL and text retrieval	2023.04
ChemCrow [45]	Ⓢ		✓	Molecular, cheminformatics, search and critique tools	2023.04

Suggest Research Ideas:

- Towards **personalized** predictions & suggestion
- Possibility to <u>suggest exciting interdisciplinary projects</u>!

A step towards an artificial scientific muse, catalyzing unforeseen collaborations and suggesting interesting research ideas for scientists